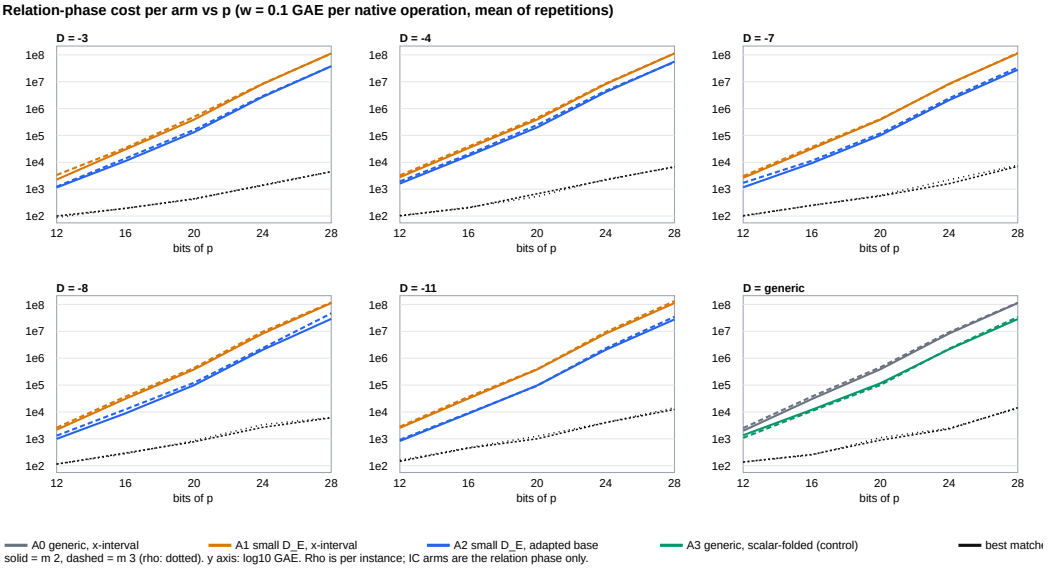
Small D_E with an endomorphism-adapted factor base

Small D_E with an endomorphism-adapted factor base: measured relation cost (2026-11-12)



Relation-phase cost per arm vs p

Protocol: PROTOCOL.md, frozen by hash before any measured run (frozen.sha256, re-verified after the run: PROTOCOL.md: OK) with Amendment 1 (pre-data) pinning cells, bases, solvers, costs and the stop rule. Code: src/cryptanalysis/chained_transfer/small_de.rs, src/bin/small_de_ic.rs; renderer render.py. Bench record: benchmarks/records/20261112T1753Z-small-de-endo-fb.json (open). Raw data: small_de.json, run.log (recorded); derived: small_de_derived.json, tables.md, small_de.svg (total relation-phase cost per arm vs p, one panel per D). Every number below is an unverified assertion produced by that run [pcc:unverified]; none is signed by pcc.

Answer

Question: after choosing coordinates and a factor base adapted to a low-degree endomorphism, does small D_E reduce the actual solving cost? No. The cost of solving one trial (the S_3 quadratic for m = 2, the S_4 quartic as two nested quadratics for m = 3) is the same, to the operation, in every arm of every cell: field operations per trial are identical for A2 and A1 (ratio 1.000) because the sweep runs over the same number of base points [pcc:unverified]. The decomposition probability per trial is unchanged for D = -3, -4 (19

of 20 cells within 2 standard errors, the other at 2.7) and is lower by 0.09 to 0.11 at `m = 2` and 20 to 28 bits for the layered `D = -7, -8` bases `[pcc:unverified]`. What does fall is the number of columns, hence the relations needed and the elimination: A2 has exactly 1/3 (`D = -3`), 1/2 (`D = -4`) or 1/4 (layered `D = -7, -8, -11`) of A1's columns, and its relation-phase cost at w = 0.1 is 0.33 to 0.35, 0.47 to 0.56 and 0.24 to 0.39 of A1's at 20 to 28 bits `[pcc:unverified]`.

That gain is the folding, not the small `D_E`. The control A3, a generic curve (`|d_K|` above 100) whose base is folded by multiplication with a random scalar in the same four layers, has relation-phase cost 0.22 to 0.30 of its x-interval twin A0 at 20 to 28 bits, and the layered `D = -7, -8, -11` arms differ from it by at most 0.097 at 24 and 28 bits (22 of 30 cells within 0.1 over all sizes) `[pcc:unverified]`. The automorphism arms (`D = -3, -4`) are beaten by the generic scalar-folded control (0.33 and 0.5 against 0.22 to 0.30), because the fold factor the control uses (4) is a free parameter while `|Aut|/2` is 3 and 2 `[pcc:unverified]`.

Against the matched rho the gap stays large and grows with `p`: no IC arm is below its matched rho at any weight in any of the 120 arm-cells; the smallest single-instance ratio is 2.32 (w = 0), 7.62 (w = 0.1) and 40.16 (w = 1); at 28 bits the amortised ratio (w = 0.1) is 125 to 1631 across arms `[pcc:unverified]`. Between 16 and 28 bits the `m = 2` amortised ratio grows by a factor of 45 (A0), 61 (A2, `D = -3`) and 106 (A1, `D = -3`), against `sqrt(2^12) =` 64, the `p` versus `sqrt(p)` gap `[pcc:unverified]`. Distance remaining, not a gate: the best row is A2 for `D = -11`, `m = 2`, 12 bits, at 2.29 times the matched rho amortised over 16 targets (w = 0.1) `[pcc:unverified]`.

Verdict against the frozen decision rule

Rule (Amendment 1): A2 is a gain for a `D` if in every `(bits, m)` cell its relation-phase total is below A1's at all three weights by more than two pooled standard errors, with every target certified and the 90% rule met.

- `D = -3, -4, -7, -8`: gain. All 10 cells of each are `win / win / win`, 16 of 16 targets certified in every arm of every cell (60 cells), the 90% rule met in every repetition `[pcc:unverified]`.
- `D = -11`: not a gain by the rule as written. The five `m = 2` cells are `loss / win / win`: at w = 0 (native work free) A2 costs more than A1 because building the layered base needs two priced group additions per image (degree-3 Velu kernel sum) and the cell's priced trial cost is small; at w = 0.1 and 1 A2 wins. The order changes inside the band, so the pair is unresolved at that boundary and not a win `[pcc:unverified]`.
- The rule passes for four `D`, but the control shows the pass is the fold factor, not `D_E` (previous section). The rule does not ask for that comparison; the control arm was added after the smoke test and is labelled so in Amendment 1.

Prediction scorecard (predictions frozen before data, in PROTOCOL.md)

1. `D = -3, -4`, P = 0.7: **partly refuted.** Columns fall by exactly 3 and 2 (holds). Decomposition probability and solving work unchanged (holds: ratio 1.000 of ops per trial; probability within 2 SE in 19 of 20 cells). The elimination falls by "about its square" (**refuted**): the ratio A1/A2 of elimination multiply-adds is 2.7 to 3.3 (`D = -3`, `m = 2`) and 2.0 to 2.1 (`D = -4`, `m = 2`), i.e. about the fold factor, because the rows stay 2 to 3 entries wide and the sparse elimination is nearly linear; at `m = 3` it is 3.1 to 7.1 and 2.0 to 3.3 against 9 and 4 predicted. The gap to the automorphism-matched rho "at least the plain-IC gap" (**refuted**): A2 over the `A = 6, 4` rho is at least A1 over plain rho in 10 of 20 cells at w = 0 and in none at w = 0.1 or 1 (A2 over the automorphism rho: minimum 2.73, 12.4, 42.3 for w = 0, 0.1, 1; A1 over plain rho: minimum 2.33, 25.3, 167) `[pcc:unverified]`.
2. `D = -7, -8, -11`, P = 0.65: **refuted.** A2 costs less than A1 in the relation phase at w = 0.1 and 1 in all 30 cells and at w = 0 in 25 of 30 (the five exceptions are the `D = -11`, `m = 2` cells) `[pcc:unverified]`. The reasoning behind the prediction (no useful base is closed under `psi` without growing by about `deg psi` per step) is right as a statement

about closure: `lambda^k = 1` for some `k <= 1000` only at 12 bits for `D = -7, -11` (orders 556 and 27) and not at all above, so no finite closed base of useful size exists; but a layered base, closed only up to its top layer, folds columns by its layer count and that is all the effect needs `[pcc:unverified]`.
3. No cell ranks A2 below the matched rho, P = 0.85: **holds**, 0 of 120 arm-cells below the best matched rho at any weight `[pcc:unverified]`.
4. Control (added after the smoke test, P = 0.8): `A3/A0` equals `A2/A1` for `D = -7, -8, -11` within noise: **holds at 24 and 28 bits** (max difference 0.097 over 12 cells), 22 of 30 cells within 0.1 over all sizes; the eight others are all at 12 and 16 bits (fixed base-build costs weigh more; the largest is `D = -11`, `m = 2`, 12 bits: 0.325 against 0.685) `[pcc:unverified]`.

Other measured facts

- The solving cost per trial depends only on the number of base points and `m`: for example 5209 field operations at 20 bits and 90546 at 28 bits for `D = -11`, `m = 2`; 50148 and 2137896 for `m = 3` (the nested quadratics), with the same value in A0, A1, A2 and A3 of a cell `[pcc:unverified]`. The degree of the univariate solved is 2 for `m = 2` and 4 (as 2 x 2) for `m = 3`; no Groebner or Macaulay matrix is built, so there is no largest matrix for the solving step, only the elimination sizes in the tables.
- The layered `D = -7, -8` bases have a lower share of trials with at least one decomposition at `m = 2` (about 0.77 and 0.76 against 0.87 at 28 bits) with the same hits per trial (2.04 and 1.95 against 2.00 and 2.02), so the hits come in clusters `[pcc:unverified]`. Cause not tested. A guess, not a result: the characteristic polynomial of `psi` makes the layers integer combinations of two of them (`psi^2 = psi - 2` for `D = -7`, `psi^2 = -2` for `D = -8`). `D = -11` (`psi^2 = psi - 3`) shows no such drop (0.869 against 0.862) `[pcc:unverified]`.
- Stop rule: the 90% rule was met in all 1416 repetitions; 562 stopped at the first checkpoint (1.0 times the columns), 786 at the second (1.5), and 68 needed 3 to 6 attempts (at most 6). Relations needed are therefore about 1.0 to 1.5 times the column count `[pcc:unverified]`.
- Matched rho recovered the log in all 16 targets for each of the 140 rho variant-cells, each certified `[k]G == T` `[pcc:unverified]`.

Tables (generated by render.py)

Curves

bits	D	p	t	#E = h*r	h	r	d_K	f(pi)	crater	end
12	-3	2131	-32	2164	4	541	-3	50	True	zeta
12	-4	2137	72	2066	2	1033	-4	29	True	iota4
12	-7	2137	-90	2228	4	557	-7	8	True	+-(1
12	-8	2179	-2	2182	2	1091	-8	33	True	sqrt-
12	-11	2137	-23	2161	1	2161	-11	27	True	+-(1
12	0	2131	79	2053	1	2053	-2283	1	None	-
16	-3	41047	-229	41277	3	13759	-3	193	True	zeta
16	-4	41057	-328	41386	2	20693	-4	119	True	iota4
16	-7	41141	106	41036	4	10259	-7	148	True	+-(1
16	-8	41227	406	40822	2	20411	-8	3	True	sqrt-
16	-11	41077	-263	41341	1	41341	-11	93	True	+-(1
16	0	41047	-115	41163	3	13721	None	None	None	-

bits	D	p	t	#E = h*r	h	r	d_K	f(pi)	crater	end
20	-3	549001	323	548679	3	182893	-3	835	True	zeta
20	-4	549121	1480	547642	2	273821	-4	39	True	iota4
20	-7	549229	-846	550076	4	137519	-7	460	True	+-(1
20	-8	549163	-1478	550642	2	275321	-8	39	True	sqrt-
20	-11	549091	-1217	550309	1	550309	-11	255	True	+-(1
20	0	548957	287	548671	1	548671	None	None	None	-
24	-3	12083299	-6907	12090207	3	4030069	-3	457	True	zeta
24	-4	12083321	928	12082394	2	6041197	-4	3445	True	iota4
24	-7	12083689	5398	12078292	4	3019573	-7	1656	True	+-(1
24	-8	12083299	2818	12080482	2	6040241	-8	2247	True	sqrt-
24	-11	12083641	6715	12076927	1	12076927	-11	543	True	+-(1
24	0	12083321	3448	12079874	2	6039937	None	None	None	-
28	-3	167272543	18827	167253717	3	55751239	-3	10241	True	zeta
28	-4	167272649	-20864	167293514	2	83646757	-4	7645	True	iota4
28	-7	167273129	-20138	167293268	4	41823317	-7	6136	True	+-(1
28	-8	167272801	-20546	167293348	4	41823337	-8	5556	True	sqrt-
28	-11	167272543	23281	167249263	1	167249263	-11	3399	True	+-(1
28	0	167272541	20725	167251817	1	167251817	None	None	None	-

D = -3: CM, End = O_K

Structure (means over repetitions). `P(dec)` = share of trials with at least one decomposition; `ops/trial` = field multiplications + inversions of the decomposition sweeps per trial; `LA` = elimination multiply-adds summed over all attempts; `rule` = repetitions that met the 90% rule; `cert` = targets whose log passed `[k]G == T`.

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	
12	2	A2	48	16	0.88	0.468	337	27.8	197	27
12	2	A1	48	48	0.87	0.471	337	71.2	525	71
12	3	A2	15	5	0.93	0.446	1710	7	32.6	7×
12	3	A1	15	15	0.82	0.495	1710	23.4	233	23
16	2	A2	204	68	0.88	0.492	1429	108	894	10
16	2	A1	204	204	0.87	0.487	1429	307	2655	30
16	3	A2	36	12	0.78	0.699	9396	14.9	123	14
16	3	A1	36	36	0.79	0.661	9396	39.2	381	39
20	2	A2	741	247	0.88	0.494	5188	372	3036	37
20	2	A1	741	741	0.88	0.487	5188	1113	10082	11
20	3	A2	84	28	0.76	0.677	50148	35.5	468	35
20	3	A1	84	84	0.77	0.701	50148	106	2216	10
24	2	A2	3480	1160	0.86	0.497	24361	1741	16273	17
24	2	A1	3480	3480	0.87	0.5	24361	5222	50861	52
24	3	A2	231	77	0.73	0.776	3.76e+05	77.5	902	77
24	3	A1	231	231	0.74	0.75	3.76e+05	232	5120	23

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	
28	2	A2	12933	4311	0.86	0.501	90532	6468	61584	64
28	2	A1	12933	12933	0.87	0.498	90532	19402	1.96e+05	19
28	3	A2	552	184	0.75	0.74	2.14e+06	184	4298	18
28	3	A1	552	552	0.76	0.733	2.14e+06	553	28195	55

Cost in GAE (priced group operations + w x native operations; relation phase includes the base build; single-instance = relation phase + mean target stage; `rho` = best matched variant, mean per target, with its name; `IC/rho` = single-instance total over that rho). Mean +- standard error of the repetitions is in `small_de_derived.json`.

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A2	505	1151	6969	982	1674	7906	96.7 (negati
12	2	A1	569	2273	17602	1049	2796	18522	96.7 (negati
12	3	A2	566	1242	7333	1073	1969	10030	86.5 (a
12	3	A1	834	3375	26245	1351	4144	29274	86.5 (a
16	2	A2	901	11095	1.03e+05	1659	12060	1.06e+05	184 (au
16	2	A1	1200	29837	2.88e+05	1970	30810	2.9e+05	184 (au
16	3	A2	1504	13929	1.26e+05	2351	16110	1.4e+05	188 (au
16	3	A1	2674	34155	3.17e+05	3538	36500	3.33e+05	188 (au
20	2	A2	1453	1.31e+05	1.3e+06	2372	1.33e+05	1.31e+06	418 (au
20	2	A1	2545	3.85e+05	3.83e+06	3463	3.86e+05	3.83e+06	418 (au
20	3	A2	5002	1.61e+05	1.57e+06	6170	1.72e+05	1.66e+06	396 (au
20	3	A1	13498	4.93e+05	4.81e+06	14608	5.01e+05	4.88e+06	396 (au
24	2	A2	3773	2.73e+06	2.72e+07	4979	2.73e+06	2.73e+07	1268 (aut)
24	2	A1	9010	8.21e+06	8.21e+07	10221	8.22e+06	8.21e+07	1268 (aut)
24	3	A2	29110	2.93e+06	2.91e+07	30926	2.99e+06	2.97e+07	1352 (aut)
24	3	A1	81800	8.47e+06	8.39e+07	83486	8.52e+06	8.44e+07	1352 (aut)
28	2	A2	10996	3.78e+07	3.78e+08	12320	3.78e+07	3.78e+08	4116 (a
28	2	A1	30328	1.13e+08	1.13e+09	31655	1.13e+08	1.13e+09	4116 (a
28	3	A2	1.52e+05	3.76e+07	3.75e+08	1.55e+05	3.81e+07	3.79e+08	4136 (aut)

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w
28	3	A1	4.49e+05	1.12e+08	1.11e+09	4.52e+05	1.12e+08	1.12e+09	4136 (aut)

Matched rho by variant (mean GAE per target; native = canonicalisations the unit does not charge; field ops = multiplications + inversions):

bits	m	variant	A	runs	GAE	native	field ops
12	2,3	plain	1	16/16	111	0	585
12	2,3	negation	2	16/16	96.7	39.1	529
12	2,3	aut	6	16/16	129	58.2	811
16	2,3	plain	1	16/16	254	0	1196
16	2,3	negation	2	16/16	221	149	1070
16	2,3	aut	6	16/16	184	77.9	1159
20	2,3	plain	1	16/16	717	0	3246
20	2,3	negation	2	16/16	595	437	2762
20	2,3	aut	6	16/16	418	241	2583
24	2,3	plain	1	16/16	2552	0	10622
24	2,3	negation	2	16/16	2967	2792	12334
24	2,3	aut	6	16/16	1268	1064	7703
28	2,3	plain	1	16/16	10617	0	43343
28	2,3	negation	2	16/16	7575	7201	31208
28	2,3	aut	6	16/16	4116	3707	24854

D = -4: CM, End = O_K

Structure (means over repetitions). `P(dec)` = share of trials with at least one decomposition; `ops/trial` = field multiplications + inversions of the decomposition sweeps per trial; `LA` = elimination multiply-adds summed over all attempts; `rule` = repetitions that met the 90% rule; `cert` = targets whose log passed `[k]G == T`.

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	I
12	2	A2	46	23	0.89	0.476	323	37.6	280	37
12	2	A1	46	46	0.87	0.504	323	71.3	556	71
12	3	A2	14	7	0.76	0.572	1497	12.8	103	12
12	3	A1	14	14	0.79	0.427	1497	31	297	31
16	2	A2	204	102	0.88	0.51	1429	158	1302	15
16	2	A1	204	204	0.87	0.509	1429	314	2781	31
16	3	A2	36	18	0.81	0.636	9396	25.1	262	25
16	3	A1	36	36	0.76	0.637	9396	48.8	582	48
20	2	A2	742	371	0.87	0.501	5195	558	4858	55
20	2	A1	742	742	0.87	0.496	5195	1114	10365	11
20	3	A2	82	41	0.67	0.816	47806	58.6	628	58
20	3	A1	82	82	0.74	0.744	47806	94.7	1600	94
24	2	A2	3476	1738	0.87	0.502	24333	2609	23999	26
24	2	A1	3476	3476	0.87	0.499	24333	5215	51032	52

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	
24	3	A2	230	115	0.75	0.702	3.72e+05	135	3335	13
24	3	A1	230	230	0.75	0.712	3.72e+05	251	6849	25
28	2	A2	12936	6468	0.87	0.497	90553	9704	93966	97
28	2	A1	12936	12936	0.86	0.5	90553	19405	1.89e+05	19
28	3	A2	552	276	0.75	0.708	2.14e+06	276	7461	27
28	3	A1	552	552	0.74	0.748	2.14e+06	553	24676	55

Cost in GAE (priced group operations + w x native operations; relation phase includes the base build; single-instance = relation phase + mean target stage; `rho` = best matched variant, mean per target, with its name; `IC/rho` = single-instance total over that rho). Mean +- standard error of the repetitions is in `small_de_derived.json`.

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A2	498	1619	11705	956	2124	12637	98.2 (negati
12	2	A1	554	2777	22780	1012	3282	23709	98.2 (negati
12	3	A2	639	1952	13768	1134	2699	16780	102 (au
12	3	A1	812	3289	25577	1306	4035	28588	102 (au
16	2	A2	925	17361	1.65e+05	1635	18291	1.68e+05	193 (au
16	2	A1	1164	33706	3.27e+05	1878	34606	3.29e+05	193 (au
16	3	A2	1798	20074	1.85e+05	2604	22538	2.02e+05	201 (au
16	3	A1	2871	38783	3.62e+05	3694	41565	3.82e+05	201 (au
20	2	A2	1736	1.99e+05	1.97e+06	2664	2.01e+05	1.98e+06	629 (au
20	2	A1	2569	3.93e+05	3.9e+06	3504	3.94e+05	3.91e+06	629 (au
20	3	A2	7394	2.48e+05	2.41e+06	8565	2.58e+05	2.5e+06	502 (au
20	3	A1	12613	4.47e+05	4.36e+06	13748	4.56e+05	4.43e+06	502 (au
24	2	A2	5050	4.14e+06	4.14e+07	6221	4.15e+06	4.14e+07	2044 (aut)
24	2	A1	8944	8.24e+06	8.23e+07	10121	8.24e+06	8.23e+07	2044 (aut)
24	3	A2	45442	4.64e+06	4.59e+07	47223	4.7e+06	4.65e+07	2178 (aut)
24	3	A1	83750	8.64e+06	8.57e+07	85470	8.7e+06	8.62e+07	2178 (aut)
28	2	A2	15841	5.64e+07	5.64e+08	17199	5.64e+07	5.64e+08	6301 (aut)

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
28	2	A1	30416	1.13e+08	1.13e+09	31779	1.13e+08	1.13e+09	6301 (aut)
28	3	A2	2.18e+05	5.4e+07	5.38e+08	2.21e+05	5.44e+07	5.42e+08	5977 (aut)
28	3	A1	4.59e+05	1.14e+08	1.14e+09	4.62e+05	1.15e+08	1.14e+09	5977 (aut)

Matched rho by variant (mean GAE per target; native = canonicalisations the unit does not charge; field ops = multiplications + inversions):

bits	m	variant	A	runs	GAE	native	field ops
12	2,3	plain	1	16/16	115	0	609
12	2,3	negation	2	16/16	98.2	38.1	542
12	2,3	aut	4	16/16	99.9	27.9	608
16	2,3	plain	1	16/16	353	0	1590
16	2,3	negation	2	16/16	274	200	1284
16	2,3	aut	4	16/16	193	104	1103
20	2,3	plain	1	16/16	799	0	3575
20	2,3	negation	2	16/16	655	493	3009
20	2,3	aut	4	16/16	629	444	3413
24	2,3	plain	1	16/16	3064	0	13107
24	2,3	negation	2	16/16	3061	2694	13106
24	2,3	aut	4	16/16	2044	1647	10753
28	2,3	plain	1	16/16	13614	0	55337
28	2,3	negation	2	16/16	9341	8963	38281
28	2,3	aut	4	16/16	6301	5887	32066

D = -7: CM, End = O_K

Structure (means over repetitions). `P(dec)` = share of trials with at least one decomposition; `ops/trial` = field multiplications + inversions of the decomposition sweeps per trial; `LA` = elimination multiply-adds summed over all attempts; `rule` = repetitions that met the 90% rule; `cert` = targets whose log passed `[k]G == T`.

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	I
12	2	A2	43	12	0.69	0.65	302	18.3	108	18
12	2	A1	43	43	0.87	0.564	302	64.8	471	64
12	3	A2	16	4	0.40	0.396	1934	11.3	52.8	11
12	3	A1	16	16	0.93	0.383	1935	22.4	199	22
16	2	A2	203	51	0.78	0.507	1422	81.7	646	81
16	2	A1	203	203	0.86	0.499	1422	306	2596	30
16	3	A2	36	9	0.79	0.639	9396	13	120	13
16	3	A1	36	36	0.81	0.639	9396	44.2	555	44
20	2	A2	744	186	0.78	0.485	5209	294	2577	29
20	2	A1	744	744	0.87	0.5	5209	1118	10224	11

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	
20	3	A2	84	21	0.81	0.663	50148	27.1	288	27
20	3	A1	84	84	0.80	0.685	50148	89.3	1180	89
24	2	A2	3476	869	0.77	0.492	24333	1307	12118	13
24	2	A1	3476	3476	0.86	0.5	24333	5215	49037	52
24	3	A2	232	58	0.65	0.779	3.79e+05	63.7	989	63
24	3	A1	232	232	0.76	0.704	3.79e+05	233	5292	23
28	2	A2	12936	3234	0.77	0.491	90553	4852	46558	48
28	2	A1	12936	12936	0.87	0.5	90553	19406	1.97e+05	19
28	3	A2	552	138	0.68	0.76	2.14e+06	163	5038	16
28	3	A1	552	552	0.71	0.776	2.14e+06	554	29137	55

Cost in GAE (priced group operations + w x native operations; relation phase includes the base build; single-instance = relation phase + mean target stage; `rho` = best matched variant, mean per target, with its name; `IC/rho` = single-instance total over that rho). Mean +- standard error of the repetitions is in `small_de_derived.json`.

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A2	538	1180	6954	1020	1705	7872	99.1 (negati
12	2	A1	567	2652	21422	1041	3170	22325	99.1 (negati
12	3	A2	628	1717	11524	1241	3307	21902	98.6 (plain)
12	3	A1	757	3041	23595	1270	3802	26586	98.6 (plain)
16	2	A2	965	9295	84260	1621	10137	86781	248 (plain)
16	2	A1	1103	31961	3.1e+05	1760	32830	3.12e+05	248 (plain)
16	3	A2	1295	11380	1.02e+05	2038	13706	1.19e+05	234 (negati
16	3	A1	2725	36969	3.45e+05	3463	38987	3.59e+05	234 (negati
20	2	A2	2055	1.01e+05	9.94e+05	2953	1.03e+05	1e+06	523 (negati
20	2	A1	2541	3.9e+05	3.88e+06	3432	3.91e+05	3.88e+06	523 (negati
20	3	A2	4007	1.2e+05	1.16e+06	5133	1.3e+05	1.25e+06	556 (negati
20	3	A1	11328	4.08e+05	3.98e+06	12420	4.16e+05	4.05e+06	556 (negati
24	2	A2	6510	2.07e+06	2.06e+07	7629	2.07e+06	2.06e+07	1479 (negati
24	2	A1	8921	8.36e+06	8.35e+07	10038	8.36e+06	8.35e+07	1479 (negati
24	3	A2	24166	2.41e+06	2.39e+07	25899	2.48e+06	2.45e+07	2037 (negati

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
24	3	A1	77735	8.09e+06	8.02e+07	79371	8.15e+06	8.07e+07	2037 (negati
28	2	A2	21483	2.79e+07	2.79e+08	22824	2.79e+07	2.79e+08	6432 (negati
28	2	A1	30429	1.13e+08	1.13e+09	31767	1.14e+08	1.13e+09	6432 (negati
28	3	A2	1.38e+05	3.4e+07	3.39e+08	1.41e+05	3.44e+07	3.42e+08	7791 (plain)
28	3	A1	4.77e+05	1.19e+08	1.18e+09	4.8e+05	1.19e+08	1.19e+09	7791 (plain)

Matched rho by variant (mean GAE per target; native = canonicalisations the unit does not charge; field ops = multiplications + inversions):

bits	m	variant	A	runs	GAE	native	field ops
12	2,3	plain	1	16/16	128	0	659
12	2,3	negation	2	16/16	99.1	39.9	540
16	2,3	plain	1	16/16	248	0	1168
16	2,3	negation	2	16/16	284	212	1323
20	2,3	plain	1	16/16	691	0	3135
20	2,3	negation	2	16/16	523	367	2465
24	2,3	plain	1	16/16	2681	0	11136
24	2,3	negation	2	16/16	1479	1305	6349
28	2,3	plain	1	16/16	11292	0	46041
28	2,3	negation	2	16/16	6432	6056	26629

D = -8: CM, End = O_K

Structure (means over repetitions). `P(dec)` = share of trials with at least one decomposition; `ops/trial` = field multiplications + inversions of the decomposition sweeps per trial; `LA` = elimination multiply-adds summed over all attempts; `rule` = repetitions that met the 90% rule; `cert` = targets whose log passed `[k]G == T`.

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	
12	2	A2	47	12	0.79	0.556	330	16.6	82.5	16
12	2	A1	47	47	0.89	0.474	330	69.2	508	69
12	3	A2	16	4	0.90	0.4	1936	8.45	40.5	8.
12	3	A1	16	16	0.96	0.335	1934	21.9	201	21
16	2	A2	203	51	0.78	0.503	1422	80.5	618	80
16	2	A1	203	203	0.87	0.491	1422	313	2773	31
16	3	A2	36	9	0.72	0.792	9396	11.5	85.3	11
16	3	A1	36	36	0.79	0.657	9396	44.5	529	44
20	2	A2	742	186	0.76	0.506	5195	280	2422	28
20	2	A1	742	742	0.86	0.498	5195	1115	10250	11
20	3	A2	84	21	0.74	0.637	50148	37.8	365	37
20	3	A1	84	84	0.74	0.669	50147	97	1498	97

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	
24	2	A2	3474	869	0.77	0.496	24319	1305	11833	13
24	2	A1	3474	3474	0.87	0.501	24319	5212	49436	52
24	3	A2	232	58	0.67	0.749	3.79e+05	64.2	968	64
24	3	A1	232	232	0.75	0.722	3.79e+05	272	10431	27
28	2	A2	12936	3234	0.76	0.514	90553	4852	47050	48
28	2	A1	12936	12936	0.87	0.496	90553	19405	2.02e+05	19
28	3	A2	552	138	0.69	0.817	2.14e+06	207	10170	20
28	3	A1	552	552	0.72	0.779	2.14e+06	553	27390	55

Cost in GAE (priced group operations + w x native operations; relation phase includes the base build; single-instance = relation phase + mean target stage; `rho` = best matched variant, mean per target, with its name; `IC/rho` = single-instance total over that rho). Mean +- standard error of the repetitions is in `small_de_derived.json`.

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A2	532	996	5172	1008	1522	6152	110 (negati
12	2	A1	567	2181	16700	1043	2707	17683	110 (negati
12	3	A2	580	1306	7843	1110	2224	12245	111 (negati
12	3	A1	721	2606	19571	1221	3324	22260	111 (negati
16	2	A2	1079	8834	78625	1859	9882	82092	275 (negati
16	2	A1	1222	30562	2.95e+05	2001	31546	2.97e+05	275 (negati
16	3	A2	1454	12317	1.1e+05	2342	15015	1.29e+05	256 (negati
16	3	A1	2851	37018	3.45e+05	3730	39466	3.61e+05	256 (negati
20	2	A2	2069	98307	9.64e+05	3001	1e+05	9.73e+05	717 (negati
20	2	A1	2576	3.78e+05	3.76e+06	3505	3.8e+05	3.77e+06	717 (negati
20	3	A2	4112	1.22e+05	1.18e+06	5316	1.34e+05	1.29e+06	789 (negati
20	3	A1	12018	4.33e+05	4.22e+06	13180	4.42e+05	4.3e+06	789 (negati
24	2	A2	6527	2.04e+06	2.03e+07	7665	2.04e+06	2.04e+07	2460 (negati
24	2	A1	8915	8.2e+06	8.19e+07	10048	8.2e+06	8.19e+07	2460 (negati
24	3	A2	23876	2.38e+06	2.36e+07	25674	2.44e+06	2.42e+07	3096 (negati
24	3	A1	92509	9.64e+06	9.56e+07	94514	9.72e+06	9.64e+07	3096 (negati

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w
28	2	A2	21574	2.92e+07	2.92e+08	22916	2.92e+07	2.92e+08	5523 (negati
28	2	A1	30329	1.13e+08	1.13e+09	31671	1.13e+08	1.13e+09	5523 (negati
28	3	A2	1.89e+05	4.67e+07	4.66e+08	1.92e+05	4.72e+07	4.7e+08	5703 (negati
28	3	A1	4.78e+05	1.19e+08	1.18e+09	4.81e+05	1.19e+08	1.19e+09	5703 (negati

Matched rho by variant (mean GAE per target; native = canonicalisations the unit does not charge; field ops = multiplications + inversions):

bits	m	variant	A	runs	GAE	native	field ops
12	2,3	plain	1	16/16	128	0	661
12	2,3	negation	2	16/16	110	47.8	586
16	2,3	plain	1	16/16	371	0	1666
16	2,3	negation	2	16/16	275	200	1292
20	2,3	plain	1	16/16	866	0	3846
20	2,3	negation	2	16/16	717	557	3255
24	2,3	plain	1	16/16	4415	0	18516
24	2,3	negation	2	16/16	2460	2092	10702
28	2,3	plain	1	16/16	11053	0	45085
28	2,3	negation	2	16/16	5523	5146	22988

D = -11: CM, End = O_K

Structure (means over repetitions). `P(dec)` = share of trials with at least one decomposition; `ops/trial` = field multiplications + inversions of the decomposition sweeps per trial; `LA` = elimination multiply-adds summed over all attempts; `rule` = repetitions that met the 90% rule; `cert` = targets whose log passed `[k]G == T`.

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	
12	2	A2	48	12	0.89	0.46	337	15.8	78.9	15
12	2	A1	48	48	0.91	0.463	337	73.8	565	73
12	3	A2	16	4	1.00	0.353	1935	7.7	49.8	7.
12	3	A1	16	16	0.93	0.369	1934	23.4	222	23
16	2	A2	204	51	0.90	0.477	1429	80.5	649	80
16	2	A1	204	204	0.87	0.495	1429	307	2676	30
16	3	A2	36	9	0.79	0.66	9396	10.3	62.5	10
16	3	A1	36	36	0.79	0.644	9396	46	464	46
20	2	A2	744	186	0.88	0.498	5209	280	2379	28
20	2	A1	744	744	0.87	0.502	5209	1117	10258	11
20	3	A2	84	21	0.77	0.651	50148	22.1	164	22
20	3	A1	84	84	0.77	0.667	50148	88.7	1201	88
24	2	A2	3473	869	0.88	0.495	24312	1304	11865	13

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	
24	2	A1	3473	3473	0.86	0.499	24312	5210	50790	52
24	3	A2	232	58	0.72	0.749	3.79e+05	63.7	964	63
24	3	A1	232	232	0.72	0.764	3.79e+05	252	7650	25
28	2	A2	12935	3234	0.87	0.492	90546	4851	45878	48
28	2	A1	12935	12935	0.86	0.499	90546	19403	1.96e+05	19
28	3	A2	552	138	0.71	0.777	2.14e+06	162	4360	16
28	3	A1	552	552	0.72	0.755	2.14e+06	645	54561	64

Cost in GAE (priced group operations + w x native operations; relation phase includes the base build; single-instance = relation phase + mean target stage; `rho` = best matched variant, mean per target, with its name; `IC/rho` = single-instance total over that rho). Mean +- standard error of the repetitions is in `small_de_derived.json`.

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A2	351	832	5158	586	1115	5880	146 (plain)
12	2	A1	334	2556	22557	567	2831	23205	146 (plain)
12	3	A2	343	923	6138	608	1435	8883	160 (plain)
12	3	A1	525	2835	23627	794	3351	26369	160 (plain)
16	2	A2	865	8761	79825	1205	9315	82304	427 (negati
16	2	A1	799	31762	3.1e+05	1138	32300	3.13e+05	427 (negati
16	3	A2	907	9140	83233	1344	11234	1e+05	443 (negati
16	3	A1	2424	36626	3.44e+05	2848	38330	3.58e+05	443 (negati
20	2	A2	2344	97047	9.49e+05	2783	98269	9.58e+05	923 (negati
20	2	A1	2115	3.83e+05	3.81e+06	2550	3.84e+05	3.82e+06	923 (negati
20	3	A2	3048	95317	9.26e+05	3705	1.04e+05	1.01e+06	1184 (negati
20	3	A1	10544	3.93e+05	3.84e+06	11230	4.03e+05	3.93e+06	1184 (negati
24	2	A2	9451	2.06e+06	2.05e+07	10006	2.07e+06	2.06e+07	3674 (negati
24	2	A1	8350	8.27e+06	8.26e+07	8901	8.28e+06	8.27e+07	3674 (negati
24	3	A2	23314	2.35e+06	2.33e+07	24506	2.42e+06	2.4e+07	3751 (negati
24	3	A1	90151	9.46e+06	9.38e+07	91296	9.52e+06	9.43e+07	3751 (negati
28	2	A2	33752	2.79e+07	2.78e+08	34400	2.79e+07	2.78e+08	11260 (negati

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w
28	2	A1	29740	1.13e+08	1.13e+09	30390	1.13e+08	1.13e+09	11260 (negati
28	3	A2	1.43e+05	3.52e+07	3.51e+08	1.45e+05	3.56e+07	3.54e+08	13040 (negati
28	3	A1	5.39e+05	1.34e+08	1.34e+09	5.41e+05	1.35e+08	1.34e+09	13040 (negati

Matched rho by variant (mean GAE per target; native = canonicalisations the unit does not charge; field ops = multiplications + inversions):

bits	m	variant	A	runs	GAE	native	field ops
12	2,3	plain	1	16/16	146	0	743
12	2,3	negation	2	16/16	178	113	876
16	2,3	plain	1	16/16	460	0	2196
16	2,3	negation	2	16/16	427	277	2066
20	2,3	plain	1	16/16	1215	0	5251
20	2,3	negation	2	16/16	923	760	4102
24	2,3	plain	1	16/16	4398	0	18458
24	2,3	negation	2	16/16	3674	3306	15578
28	2,3	plain	1	16/16	16360	0	66330
28	2,3	negation	2	16/16	11260	10877	45975

D = 0: generic (D=0, large |d_K|)

Structure (means over repetitions). `P(dec)` = share of trials with at least one decomposition; `ops/trial` = field multiplications + inversions of the decomposition sweeps per trial; `LA` = elimination multiply-adds summed over all attempts; `rule` = repetitions that met the 90% rule; `cert` = targets whose log passed `[k]G == T`.

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	
12	2	A3	47	12	0.94	0.446	330	17.7	100	17
12	2	A0	47	47	0.92	0.468	330	72.7	555	72
12	3	A3	16	4	0.95	0.428	1936	5.15	22.5	5.
12	3	A0	16	16	0.95	0.336	1936	25.5	221	25
16	2	A3	204	51	0.87	0.508	1429	80.2	640	80
16	2	A0	204	204	0.87	0.494	1429	307	2610	30
16	3	A3	36	9	0.84	0.641	9396	11.9	93.9	11
16	3	A0	36	36	0.78	0.628	9396	50	567	50
20	2	A3	744	186	0.86	0.498	5209	280	2405	28
20	2	A0	744	744	0.86	0.501	5209	1117	10496	11
20	3	A3	84	21	0.80	0.674	50148	21.8	150	21
20	3	A0	84	84	0.74	0.748	50148	93.2	1481	93
24	2	A3	3475	869	0.76	0.497	24326	1305	12146	13
24	2	A0	3475	3475	0.86	0.503	24326	5214	49891	52
24	3	A3	232	58	0.72	0.77	3.79e+05	59.3	695	59

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	I
24	3	A0	232	232	0.74	0.759	3.79e+05	252	7315	25
28	2	A3	12934	3234	0.87	0.494	90539	4853	46002	48
28	2	A0	12934	12934	0.87	0.498	90539	19402	1.93e+05	19
28	3	A3	552	138	0.71	0.761	2.14e+06	161	4761	16
28	3	A0	552	552	0.74	0.76	2.14e+06	552	33674	55

Cost in GAE (priced group operations + w x native operations; relation phase includes the base build; single-instance = relation phase + mean target stage; `rho` = best matched variant, mean per target, with its name; `IC/rho` = single-instance total over that rho). Mean +- standard error of the repetitions is in `small_de_derived.json`.

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A3	972	1358	4837	1200	1637	5569	131 (negati
12	2	A0	332	1983	16841	563	2259	17528	131 (negati
12	3	A3	543	1088	5991	816	1640	9054	129 (negati
12	3	A0	513	2509	20476	783	3043	23382	129 (negati
16	2	A3	4282	12044	81899	4990	12942	84511	242 (negati
16	2	A0	1147	30206	2.92e+05	1858	31126	2.95e+05	242 (negati
16	3	A3	1782	10817	92126	2614	13608	1.13e+05	242 (negati
16	3	A0	2828	37480	3.49e+05	3645	39949	3.67e+05	242 (negati
20	2	A3	20199	1.17e+05	9.87e+05	20635	1.18e+05	9.93e+05	824 (negati
20	2	A0	2115	3.9e+05	3.88e+06	2554	3.91e+05	3.89e+06	824 (negati
20	3	A3	5122	1e+05	9.56e+05	5800	1.1e+05	1.04e+06	1018 (negati
20	3	A0	12283	4.62e+05	4.51e+06	12914	4.7e+05	4.58e+06	1018 (negati
24	2	A3	1.04e+05	2.15e+06	2.06e+07	1.05e+05	2.16e+06	2.06e+07	2150 (negati
24	2	A0	8904	8.3e+06	8.29e+07	10007	8.31e+06	8.3e+07	2150 (negati
24	3	A3	28164	2.25e+06	2.23e+07	29896	2.32e+06	2.29e+07	2310 (negati
24	3	A0	89598	9.34e+06	9.26e+07	91273	9.4e+06	9.32e+07	2310 (negati
28	2	A3	5.25e+05	2.85e+07	2.81e+08	5.26e+05	2.86e+07	2.81e+08	13055 (negati
28	2	A0	29700	1.13e+08	1.13e+09	30352	1.13e+08	1.13e+09	13055 (negati

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
28	3	A3	1.57e+05	3.43e+07	3.42e+08	1.59e+05	3.46e+07	3.45e+08	12187 (negati
28	3	A0	4.65e+05	1.16e+08	1.15e+09	4.67e+05	1.16e+08	1.16e+09	12187 (negati

Matched rho by variant (mean GAE per target; native = canonicalisations the unit does not charge; field ops = multiplications + inversions):

bits	m	variant	A	runs	GAE	native	field ops
12	2,3	plain	1	16/16	139	0	706
12	2,3	negation	2	16/16	131	69.1	678
16	2,3	plain	1	16/16	280	0	1298
16	2,3	negation	2	16/16	242	169	1157
20	2,3	plain	1	16/16	1247	0	5377
20	2,3	negation	2	16/16	824	659	3693
24	2,3	plain	1	16/16	3628	0	15361
24	2,3	negation	2	16/16	2150	1784	9455
28	2,3	plain	1	16/16	19190	0	77654
28	2,3	negation	2	16/16	13055	12676	53172

Verdict against the frozen decision rule (A2 vs A1, relation phase)

`win` = A2 below A1 by more than 2 pooled standard errors, `tie` = within, `loss` = above. One entry per weight (0 / 0.1 / 1). A D is a gain only if every cell is `win` at all three weights, every target certified, and the 90% rule met in all repetitions.

D	bits	m	rel-phase (w=0 / 0.1 / 1)	single-instance	A2/A1 rel ratio (w=0.1)	columns A2/A1	LA A2/A1	rule A2,A1	cert A2,A1
-3	12	2	win / win / win	win / win / win	0.506	16/48	0.375	20/20,20/20	16/16,16/16
-3	12	3	win / win / win	win / win / win	0.368	5/15	0.140	20/20,20/20	16/16,16/16
-3	16	2	win / win / win	win / win / win	0.372	68/204	0.337	20/20,20/20	16/16,16/16
-3	16	3	win / win / win	win / win / win	0.408	12/36	0.323	20/20,20/20	16/16,16/16
-3	20	2	win / win / win	win / win / win	0.341	247/741	0.301	10/10,10/10	16/16,16/16
-3	20	3	win / win / win	win / win / win	0.327	28/84	0.211	10/10,10/10	16/16,16/16

D	bits	m	rel-phase (w=0 / 0.1 / 1)	single-instance	A2/A1 rel ratio (w=0.1)	columns A2/A1	LA A2/A1	rule A2,A1	cert A2,A1
-3	24	2	win / win / win	win / win / win	0.332	1160/3480	0.320	6/6,6/6	16/16,16/16
-3	24	3	win / win / win	win / win / win	0.347	77/231	0.176	6/6,6/6	16/16,16/16
-3	28	2	win / win / win	win / win / win	0.336	4311/12933	0.315	3/3,3/3	16/16,16/16
-3	28	3	win / win / win	win / win / win	0.337	184/552	0.152	3/3,3/3	16/16,16/16
-4	12	2	win / win / win	win / win / win	0.583	23/46	0.504	20/20,20/20	16/16,16/16
-4	12	3	win / win / win	win / win / win	0.594	7/14	0.346	20/20,20/20	16/16,16/16
-4	16	2	win / win / win	win / win / win	0.515	102/204	0.468	20/20,20/20	16/16,16/16
-4	16	3	win / win / win	win / win / win	0.518	18/36	0.451	20/20,20/20	16/16,16/16
-4	20	2	win / win / win	win / win / win	0.507	371/742	0.469	10/10,10/10	16/16,16/16
-4	20	3	win / win / win	win / win / win	0.555	41/82	0.393	10/10,10/10	16/16,16/16
-4	24	2	win / win / win	win / win / win	0.503	1738/3476	0.470	6/6,6/6	16/16,16/16
-4	24	3	win / win / win	win / win / win	0.536	115/230	0.487	6/6,6/6	16/16,16/16
-4	28	2	win / win / win	win / win / win	0.498	6468/12936	0.496	3/3,3/3	16/16,16/16
-4	28	3	win / win / win	win / win / win	0.473	276/552	0.302	3/3,3/3	16/16,16/16
-7	12	2	win / win / win	win / win / win	0.445	12/43	0.229	20/20,20/20	16/16,16/16
-7	12	3	win / win / win	tie / tie / tie	0.565	4/16	0.264	20/20,20/20	16/16,16/16
-7	16	2	win / win /	win / win / win	0.291	51/203	0.249	20/20,20/20	16/16,16/16

D	bits	m	rel-phase (w=0 / 0.1 / 1)	single-instance	A2/A1 rel ratio (w=0.1)	columns A2/A1	LA A2/A1	rule A2,A1	cert A2,A1
			win						
-7	16	3	win / win / win	win / win / win	0.308	9/36	0.216	20/20,20/20	16/16,16/16
-7	20	2	win / win / win	win / win / win	0.260	186/744	0.252	10/10,10/10	16/16,16/16
-7	20	3	win / win / win	win / win / win	0.294	21/84	0.244	10/10,10/10	16/16,16/16
-7	24	2	win / win / win	win / win / win	0.247	869/3476	0.247	6/6,6/6	16/16,16/16
-7	24	3	win / win / win	win / win / win	0.298	58/232	0.187	6/6,6/6	16/16,16/16
-7	28	2	win / win / win	win / win / win	0.246	3234/12936	0.236	3/3,3/3	16/16,16/16
-7	28	3	win / win / win	win / win / win	0.287	138/552	0.173	3/3,3/3	16/16,16/16
-8	12	2	win / win / win	win / win / win	0.457	12/47	0.162	20/20,20/20	16/16,16/16
-8	12	3	win / win / win	win / win / win	0.501	4/16	0.202	20/20,20/20	16/16,16/16
-8	16	2	win / win / win	win / win / win	0.289	51/203	0.223	20/20,20/20	16/16,16/16
-8	16	3	win / win / win	win / win / win	0.333	9/36	0.161	20/20,20/20	16/16,16/16
-8	20	2	win / win / win	win / win / win	0.260	186/742	0.236	10/10,10/10	16/16,16/16
-8	20	3	win / win / win	win / win / win	0.282	21/84	0.244	10/10,10/10	16/16,16/16
-8	24	2	win / win / win	win / win / win	0.248	869/3474	0.239	6/6,6/6	16/16,16/16
-8	24	3	win / win / win	win / win / win	0.247	58/232	0.093	6/6,6/6	16/16,16/16
-8	28	2	win / win / win	win / win / win	0.259	3234/12936	0.233	3/3,3/3	16/16,16/16

D	bits	m	rel-phase (w=0 / 0.1 / 1)	single-instance	A2/A1 rel ratio (w=0.1)	columns A2/A1	LA A2/A1	rule A2,A1	cert A2,A1
-8	28	3	win / win / win	win / win / win	0.393	138/552	0.371	3/3,3/3	16/16,16/16
-11	12	2	loss / win / win	loss / win / win	0.325	12/48	0.140	20/20,20/20	16/16,16/16
-11	12	3	win / win / win	win / win / win	0.325	4/16	0.224	20/20,20/20	16/16,16/16
-11	16	2	loss / win / win	loss / win / win	0.276	51/204	0.242	20/20,20/20	16/16,16/16
-11	16	3	win / win / win	win / win / win	0.250	9/36	0.135	20/20,20/20	16/16,16/16
-11	20	2	loss / win / win	loss / win / win	0.253	186/744	0.232	10/10,10/10	16/16,16/16
-11	20	3	win / win / win	win / win / win	0.242	21/84	0.136	10/10,10/10	16/16,16/16
-11	24	2	loss / win / win	loss / win / win	0.249	869/3473	0.234	6/6,6/6	16/16,16/16
-11	24	3	win / win / win	win / win / win	0.249	58/232	0.126	6/6,6/6	16/16,16/16
-11	28	2	loss / win / win	loss / win / win	0.247	3234/12935	0.234	3/3,3/3	16/16,16/16
-11	28	3	win / win / win	win / win / win	0.262	138/552	0.080	3/3,3/3	16/16,16/16

Per D: D=-3: gain; D=-4: gain; D=-7: gain; D=-8: gain; D=-11: not a gain in every cell

Control: folding without a small D_E (A3 vs A0 on the generic curve)

bits	m	A3/A0 rel (w=0.1)	A3 vs A0	columns A3/A0	LA A3/A0
12	2	0.685	loss/win/win	12/47	0.181
12	3	0.434	tie/win/win	4/16	0.102
16	2	0.399	loss/win/win	51/204	0.245
16	3	0.289	win/win/win	9/36	0.166
20	2	0.300	loss/win/win	186/744	0.229
20	3	0.217	win/win/win	21/84	0.101
24	2	0.259	loss/win/win	869/3475	0.243
24	3	0.241	win/win/win	58/232	0.095

bits	m	A3/A0 rel (w=0.1)	A3 vs A0	columns A3/A0	LA A3/A0
28	2	0.253	loss/win/win	3234/12934	0.238
28	3	0.296	win/win/win	138/552	0.141

A2/A1 minus A3/A0 for the layered D (relation phase, w=0.1; a negative number means the endomorphism arm is cheaper than the generic scalar-folded arm):

D	bits	m	A2/A1	A3/A0	difference
-7	12	2	0.445	0.685	-0.240
-7	12	3	0.565	0.434	+0.131
-7	16	2	0.291	0.399	-0.108
-7	16	3	0.308	0.289	+0.019
-7	20	2	0.260	0.300	-0.040
-7	20	3	0.294	0.217	+0.077
-7	24	2	0.247	0.259	-0.012
-7	24	3	0.298	0.241	+0.057
-7	28	2	0.246	0.253	-0.008
-7	28	3	0.287	0.296	-0.009
-8	12	2	0.457	0.685	-0.228
-8	12	3	0.501	0.434	+0.068
-8	16	2	0.289	0.399	-0.110
-8	16	3	0.333	0.289	+0.044
-8	20	2	0.260	0.300	-0.040
-8	20	3	0.282	0.217	+0.066
-8	24	2	0.248	0.259	-0.011
-8	24	3	0.247	0.241	+0.005
-8	28	2	0.259	0.253	+0.006
-8	28	3	0.393	0.296	+0.097
-11	12	2	0.325	0.685	-0.360
-11	12	3	0.325	0.434	-0.108
-11	16	2	0.276	0.399	-0.123
-11	16	3	0.250	0.289	-0.039
-11	20	2	0.253	0.300	-0.046
-11	20	3	0.242	0.217	+0.026
-11	24	2	0.249	0.259	-0.010
-11	24	3	0.249	0.241	+0.008
-11	28	2	0.247	0.253	-0.006
-11	28	3	0.262	0.296	-0.034

Prediction scorecard

Prediction 1 (D = -3, -4): columns /|Aut|/2, LA about its square, P(dec) and solving unchanged, gap to the automorphism rho at least the plain gap

D	bits	m	cols A1/A2 (pred)	LA A1/A2 (pred = square)	P(dec) diff (SE)	ops/trial A2/A1	gap A2/rho_aut >= A1/rho_plain at w=0,0.1,1 (values)
-3	12	2	3.00 (3)	2.7 (9)	0.018 (0.024)	1.000	False (7.6>=9.5, 12.4>=25.3, 42.3>=167.3)
-3	12	3	3.00 (3)	7.1 (9)	0.113 (0.042)	1.000	False (12.4>=13.5, 22.3>=41.4, 96.6>=292.4)
-3	16	2	3.00 (3)	3.0 (9)	0.006 (0.011)	1.000	False (9.0>=7.7, 63.0>=121.1, 403.9>=1140.9)
-3	16	3	3.00 (3)	3.1 (9)	0.010 (0.033)	1.000	False (12.5>=12.1, 82.0>=124.5, 518.4>=1136.6)
-3	20	2	3.00 (3)	3.3 (9)	0.001 (0.011)	1.000	False (5.7>=4.8, 300.5>=538.7, 1984.3>=5343.5)
-3	20	3	3.00 (3)	4.7 (9)	0.011 (0.035)	1.000	False (15.6>=17.7, 411.2>=607.7, 2704.8>=5918.1)
-3	24	2	3.00 (3)	3.1 (9)	0.003 (0.005)	1.000	False (3.9>=4.0, 1985.4>=3220.9, 11685.4>=32173.0)
-3	24	3	3.00 (3)	5.7 (9)	0.011 (0.033)	1.000	False (22.9>=30.8, 2041.4>=3137.9, 11858.9>=31102.1)
-3	28	2	3.00 (3)	3.2 (9)	0.002 (0.003)	1.000	False (3.0>=3.0, 8428.6>=10603.0, 48325.4>=106003.4)
-3	28	3	3.00 (3)	6.6 (9)	0.009 (0.020)	1.000	False (37.5>=40.0, 8440.4>=9908.9, 48228.6>=98729.3)
-4	12	2	2.00 (2)	2.0 (4)	0.024 (0.020)	1.000	False (9.6>=8.8, 20.7>=28.5, 98.9>=205.6)
-4	12	3	2.00 (2)	2.9 (4)	0.033 (0.050)	1.000	False (11.2>=10.6, 25.8>=32.7, 128.4>=231.5)
-4	16	2	2.00 (2)	2.1 (4)	0.004 (0.009)	1.000	False (8.5>=5.3, 90.0>=98.1, 566.5>=932.6)
-4	16	3	2.00 (2)	2.2 (4)	0.047 (0.036)	1.000	False (12.9>=9.5, 106.0>=106.8, 643.4>=982.6)
-4	20	2	2.00 (2)	2.1 (4)	0.004 (0.009)	1.000	False (4.2>=4.4, 297.9>=493.5, 1847.8>=4895.8)

D	bits	m	cols A1/A2 (pred)	LA A1/A2 (pred = square)	P(dec) diff (SE)	ops/trial A2/A1	gap A2/rho_aut >= A1/rho_plain at w=0,0.1,1 (values)
-4	20	3	2.00 (2)	2.5 (4)	0.069 (0.039)	1.000	False (17.1>=15.4, 483.8>=511.3, 3066.5>=4974.3)
-4	24	2	2.00 (2)	2.1 (4)	0.002 (0.004)	1.000	False (3.0>=3.3, 1878.1>=2689.8, 11224.2>=26867.8)
-4	24	3	2.00 (2)	2.1 (4)	0.003 (0.012)	1.000	False (21.7>=22.6, 1994.3>=2296.7, 11768.7>=22763.4)
-4	28	2	2.00 (2)	2.0 (4)	0.007 (0.006)	1.000	False (2.7>=2.3, 8183.4>=8322.5, 46247.4>=83203.6)
-4	28	3	2.00 (2)	3.3 (4)	0.017 (0.027)	1.000	False (37.0>=39.7, 8324.1>=9855.9, 46946.8>=98200.9)

Prediction 2 (D = -7, -8, -11: A2 costs no less than A1)

Cells where A2 wins at all three weights: 25 of 30; cells where A2 loses at some weight: 5.

Prediction 3 (no cell ranks an IC arm below its matched rho across the band)

Arm-cells with single-instance total below the best matched rho at all three weights: 0 of 120; at some weight: 0; the smallest IC/rho ratio per weight (w=0, 0.1, 1): 2.32, 7.62, 40.16.

Distance to the matched rho (single-instance IC total / best matched rho), minimum over cells per arm

arm	w=0	w=0.1	w=1	cell of the minimum at w=0.1
A0	2.32	16.43	126.50	D=0 b=12 m=2
A1	2.42	19.33	137.88	D=-11 b=12 m=2
A2	2.72	7.62	40.16	D=-11 b=12 m=2
A3	5.70	11.91	40.19	D=0 b=12 m=2

Compact views

Amortised over the 16 targets (relation phase / 16 + mean target stage), w = 0.1, divided by the best matched rho per target:

D	arm	m	12 bits	16 bits	20 bits	24 bits	28 bits
-3	A1	2	6.61	14.82	58.07	377.10	1571.29
-3	A1	3	11.10	22.80	94.17	393.56	1631.23
-3	A2	2	5.92	8.66	22.40	127.17	529.64
-3	A2	3	9.11	15.53	49.98	165.43	616.62
-4	A1	2	6.65	14.80	39.00	235.17	1029.51
-4	A1	3	9.10	24.49	68.24	252.49	1146.49
-4	A2	2	5.95	9.92	20.95	119.51	513.23

D	arm	m	12 bits	16 bits	20 bits	24 bits	28 bits
-4	A2	3	8.30	17.50	47.71	149.33	574.87
-7	A1	2	6.63	11.53	46.26	327.62	1009.89
-7	A1	3	9.65	17.29	56.60	251.58	1003.96
-7	A2	2	5.81	5.73	14.58	82.87	250.28
-7	A2	3	17.22	12.13	29.33	96.69	313.87
-8	A1	2	5.80	9.83	32.71	193.86	1170.00
-8	A1	3	7.78	17.36	42.75	204.13	1264.29
-8	A2	2	5.15	5.43	10.21	50.17	304.70
-8	A2	3	8.82	12.64	22.68	63.53	535.03
-11	A1	2	2.97	5.56	25.16	130.18	572.99
-11	A1	3	4.35	8.45	27.28	158.88	620.67
-11	A2	2	2.29	2.42	7.29	33.19	142.15
-11	A2	3	3.58	5.64	12.03	51.20	179.64
0	A0	2	2.91	10.83	28.68	224.65	492.90
0	A0	3	5.09	19.40	33.06	257.16	569.05
0	A3	2	2.64	6.37	9.40	60.12	125.45
0	A3	3	4.57	13.98	14.24	81.30	182.81

Secondary, labelled: field operations of the relation phase (multiplications + inversions + elimination multiply-adds + 4 per priced group operation; membership tests excluded) over the field operations of the best matched rho per target (its exact tally):

D	arm	m	12 bits	16 bits	20 bits	24 bits	28 bits
-3	A1	2	30	215	1173	8298	35230
-3	A1	3	48	216	1546	7991	34913
-3	A2	2	14	78	399	2753	11831
-3	A2	3	15	87	505	2769	11753
-4	A1	2	39	240	1027	5972	27502
-4	A1	3	38	255	1238	5860	29117
-4	A2	2	21	122	520	3004	13685
-4	A2	3	22	131	686	3143	13772
-7	A1	2	37	215	1238	10299	33179
-7	A1	3	40	248	1204	7295	28729
-7	A2	2	14	60	319	2543	8147
-7	A2	3	21	75	353	2175	8247
-8	A1	2	26	183	905	5961	38202
-8	A1	3	30	228	936	5638	38919
-8	A2	2	10	51	233	1480	9885
-8	A2	3	14	74	264	1390	15301
-11	A1	2	27	121	728	4147	19113
-11	A1	3	26	130	588	4612	19636
-11	A2	2	7	32	183	1032	4713
-11	A2	3	8	32	142	1148	5151
0	A0	2	22	202	827	6847	16502

D	arm	m	12 bits	16 bits	20 bits	24 bits	28 bits
0	A0	3	26	244	793	7161	18095
0	A3	2	10	68	227	1735	4141
0	A3	3	10	67	170	1725	5362

Limitations

- One curve per `(bits, D)`; the variance across curves with the same `D` is not measured, and the generic curve differs from the CM curves, so A0 and A1 are different groups. Repetitions vary only the trial walk and jumps `[pcc:unverified]`.
- The fold factor of the layered bases and the control is fixed at 4 (Amendment 1). The effect scales with the number of layers: nothing measured here says what happens at other values. By the same accounting a base folded by a scalar could be folded far deeper on any curve; as a derivation, not a measurement, the sweep work per relation then tends to a BSGS-like cost. Not run.
- The native work is priced only through the weights 0, 0.1 and 1; at w = 0 the decomposition sweeps are free and IC arms are compared on priced operations alone, at w = 1 every field multiplication costs a group addition. Nothing here is a calibrated wall-time claim. The secondary field-operation view (compact views, last table) is labelled and excludes membership tests `[pcc:unverified]`.
- `psi` evaluation is charged analytically (kernel size minus one group additions plus two multiplications per image), not by counted operations in the Velu code. The control's base build uses counted scalar multiplications (about 36 group operations per image), which is why A3 loses to A0 at w = 0.
- Membership in the adapted base is one hash probe on `x`, not on the invariant `u = x^3` or `u = x^2` (no computation of `u` is charged; this can only favour A2 by one or two multiplications per test).
- Targets are solved from the first repetition's known columns only; a target needs a decomposition into known points, so the descent depends on that repetition's determined set (at least 90% of the columns).
- Sizes are 12 to 28 bits (the 28-bit tier is an addition to the 12 to 24 bits of the task, within the protocol's "as large as the solver allows"); `r` is about `p/h` with `h` at most 4.
- The linear algebra is sparse modular elimination, counted as multiply-adds; no block Wiedemann or Lanczos is priced.
- Curve construction (one-time, a few milliseconds per curve up to 28 bits, logged in `run.log`) is excluded from every total, as the protocol says.

Deviations from the task text

- Added 28 bits and the control arm A3, both pinned in Amendment 1 before data.
- Matched rho for `D = -7, -8, -11` and generic is negation (`A = 2`) and plain, as no automorphism group is usable; for `D = -3, -4` the group runs with `A = 6, 4` and its verification ledger is charged.
- No Groebner or Macaulay step for `m = 3` (nested quadratics instead); "largest matrix" is not applicable to the solving step.
- `benchmarks/progress/*` was regenerated by `benchrec report`, as the benchmark-record rule requires; it may conflict with other open pull requests that also regenerate it.

Reproduce

```
cargo build --release --bin small_de_ic
python3 tools/benchrec.py run $BENCHREC -- ./target/release/small_de_ic \
  --sizes 12,16,20,24,28 --reps 12:20,16:20,20:10,24:6,28:3 --targets 16 \
  --out-dir research/small_de_endo_fb_20261112
```

```
python3 research/small_de_endo_fb_20261112/render.py
research/small_de_endo_fb_20261112/small_de.json
```

The run took 1 minute 44 seconds on the shared 4-core host (`[pcc:unverified]`). Tests: `cargo test --release --lib chained_transfer::small_de` (7 pass: `lambda_action_each_d`, `aut_base_is_closed_and_logs_transport`, `layered_base_transports_logs_for_psi_and_scalar`, `decompositions_verify_and_logs_certify`, `s3_quadratic_roots_are_the_sum_and_difference`, `sparse_solver_matches_truth`, `generic_curve_has_large_dk`); `cargo clippy --release --all-targets -- -D warnings` clean for the whole crate.

Tables

Curves

bits	D	p	t	#E = h*r	h	r	d_K	f(pi)	crater	endo
12	-3	2131	-32	2164	4	541	-3	50	True	zeta
12	-4	2137	72	2066	2	1033	-4	29	True	iota4
12	-7	2137	-90	2228	4	557	-7	8	True	+-(1+
12	-8	2179	-2	2182	2	1091	-8	33	True	sqrt-
12	-11	2137	-23	2161	1	2161	-11	27	True	+-(1+
12	0	2131	79	2053	1	2053	-2283	1	None	-
16	-3	41047	-229	41277	3	13759	-3	193	True	zeta
16	-4	41057	-328	41386	2	20693	-4	119	True	iota4
16	-7	41141	106	41036	4	10259	-7	148	True	+-(1+
16	-8	41227	406	40822	2	20411	-8	3	True	sqrt-
16	-11	41077	-263	41341	1	41341	-11	93	True	+-(1+
16	0	41047	-115	41163	3	13721	None	None	None	-
20	-3	549001	323	548679	3	182893	-3	835	True	zeta
20	-4	549121	1480	547642	2	273821	-4	39	True	iota4
20	-7	549229	-846	550076	4	137519	-7	460	True	+-(1+
20	-8	549163	-1478	550642	2	275321	-8	39	True	sqrt-
20	-11	549091	-1217	550309	1	550309	-11	255	True	+-(1+
20	0	548957	287	548671	1	548671	None	None	None	-
24	-3	12083299	-6907	12090207	3	4030069	-3	457	True	zeta
24	-4	12083321	928	12082394	2	6041197	-4	3445	True	iota4
24	-7	12083689	5398	12078292	4	3019573	-7	1656	True	+-(1+
24	-8	12083299	2818	12080482	2	6040241	-8	2247	True	sqrt-
24	-11	12083641	6715	12076927	1	12076927	-11	543	True	+-(1+
24	0	12083321	3448	12079874	2	6039937	None	None	None	-
28	-3	167272543	18827	167253717	3	55751239	-3	10241	True	zeta
28	-4	167272649	-20864	167293514	2	83646757	-4	7645	True	iota4
28	-7	167273129	-20138	167293268	4	41823317	-7	6136	True	+-(1+
28	-8	167272801	-20546	167293348	4	41823337	-8	5556	True	sqrt-

bits	D	p	t	#E = h*r	h	r	d_K	f(pi)	crater	end
28	-11	167272543	23281	167249263	1	167249263	-11	3399	True	+-(1
28	0	167272541	20725	167251817	1	167251817	None	None	None	-

D = -3: CM, End = O_K

Structure (means over repetitions). `P(dec)` = share of trials with at least one decomposition; `ops/trial` = field multiplications + inversions of the decomposition sweeps per trial; `LA` = elimination multiply-adds summed over all attempts; `rule` = repetitions that met the 90% rule; `cert` = targets whose log passed `[k]G == T`.

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	
12	2	A2	48	16	0.88	0.468	337	27.8	197	27
12	2	A1	48	48	0.87	0.471	337	71.2	525	71
12	3	A2	15	5	0.93	0.446	1710	7	32.6	7×
12	3	A1	15	15	0.82	0.495	1710	23.4	233	23
16	2	A2	204	68	0.88	0.492	1429	108	894	10
16	2	A1	204	204	0.87	0.487	1429	307	2655	30
16	3	A2	36	12	0.78	0.699	9396	14.9	123	14
16	3	A1	36	36	0.79	0.661	9396	39.2	381	39
20	2	A2	741	247	0.88	0.494	5188	372	3036	37
20	2	A1	741	741	0.88	0.487	5188	1113	10082	11
20	3	A2	84	28	0.76	0.677	50148	35.5	468	35
20	3	A1	84	84	0.77	0.701	50148	106	2216	10
24	2	A2	3480	1160	0.86	0.497	24361	1741	16273	17
24	2	A1	3480	3480	0.87	0.5	24361	5222	50861	52
24	3	A2	231	77	0.73	0.776	3.76e+05	77.5	902	77
24	3	A1	231	231	0.74	0.75	3.76e+05	232	5120	23
28	2	A2	12933	4311	0.86	0.501	90532	6468	61584	64
28	2	A1	12933	12933	0.87	0.498	90532	19402	1.96e+05	19
28	3	A2	552	184	0.75	0.74	2.14e+06	184	4298	18
28	3	A1	552	552	0.76	0.733	2.14e+06	553	28195	55

Cost in GAE (priced group operations + w x native operations; relation phase includes the base build; single-instance = relation phase + mean target stage; `rho` = best matched variant, mean per target, with its name; `IC/rho` = single-instance total over that rho). Mean +- standard error of the repetitions is in `small_de_derived.json`.

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A2	505	1151	6969	982	1674	7906	96.7 (negati
12	2	A1	569	2273	17602	1049	2796	18522	96.7 (negati
12	3	A2	566	1242	7333	1073	1969	10030	86.5 (a

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	3	A1	834	3375	26245	1351	4144	29274	86.5 (a
16	2	A2	901	11095	1.03e+05	1659	12060	1.06e+05	184 (au
16	2	A1	1200	29837	2.88e+05	1970	30810	2.9e+05	184 (au
16	3	A2	1504	13929	1.26e+05	2351	16110	1.4e+05	188 (au
16	3	A1	2674	34155	3.17e+05	3538	36500	3.33e+05	188 (au
20	2	A2	1453	1.31e+05	1.3e+06	2372	1.33e+05	1.31e+06	418 (au
20	2	A1	2545	3.85e+05	3.83e+06	3463	3.86e+05	3.83e+06	418 (au
20	3	A2	5002	1.61e+05	1.57e+06	6170	1.72e+05	1.66e+06	396 (au
20	3	A1	13498	4.93e+05	4.81e+06	14608	5.01e+05	4.88e+06	396 (au
24	2	A2	3773	2.73e+06	2.72e+07	4979	2.73e+06	2.73e+07	1268 (aut)
24	2	A1	9010	8.21e+06	8.21e+07	10221	8.22e+06	8.21e+07	1268 (aut)
24	3	A2	29110	2.93e+06	2.91e+07	30926	2.99e+06	2.97e+07	1352 (aut)
24	3	A1	81800	8.47e+06	8.39e+07	83486	8.52e+06	8.44e+07	1352 (aut)
28	2	A2	10996	3.78e+07	3.78e+08	12320	3.78e+07	3.78e+08	4116 (a
28	2	A1	30328	1.13e+08	1.13e+09	31655	1.13e+08	1.13e+09	4116 (a
28	3	A2	1.52e+05	3.76e+07	3.75e+08	1.55e+05	3.81e+07	3.79e+08	4136 (aut)
28	3	A1	4.49e+05	1.12e+08	1.11e+09	4.52e+05	1.12e+08	1.12e+09	4136 (aut)

Matched rho by variant (mean GAE per target; native = canonicalisations the unit does not charge; field ops = multiplications + inversions):

bits	m	variant	A	runs	GAE	native	field ops
12	2,3	plain	1	16/16	111	0	585
12	2,3	negation	2	16/16	96.7	39.1	529
12	2,3	aut	6	16/16	129	58.2	811
16	2,3	plain	1	16/16	254	0	1196
16	2,3	negation	2	16/16	221	149	1070
16	2,3	aut	6	16/16	184	77.9	1159
20	2,3	plain	1	16/16	717	0	3246
20	2,3	negation	2	16/16	595	437	2762
20	2,3	aut	6	16/16	418	241	2583

bits	m	variant	A	runs	GAE	native	field ops
24	2,3	plain	1	16/16	2552	0	10622
24	2,3	negation	2	16/16	2967	2792	12334
24	2,3	aut	6	16/16	1268	1064	7703
28	2,3	plain	1	16/16	10617	0	43343
28	2,3	negation	2	16/16	7575	7201	31208
28	2,3	aut	6	16/16	4116	3707	24854

D = -4: CM, End = O_K

Structure (means over repetitions). `P(dec)` = share of trials with at least one decomposition; `ops/trial` = field multiplications + inversions of the decomposition sweeps per trial; `LA` = elimination multiply-adds summed over all attempts; `rule` = repetitions that met the 90% rule; `cert` = targets whose log passed `[k]G == T`.

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	
12	2	A2	46	23	0.89	0.476	323	37.6	280	37
12	2	A1	46	46	0.87	0.504	323	71.3	556	71
12	3	A2	14	7	0.76	0.572	1497	12.8	103	12
12	3	A1	14	14	0.79	0.427	1497	31	297	31
16	2	A2	204	102	0.88	0.51	1429	158	1302	15
16	2	A1	204	204	0.87	0.509	1429	314	2781	31
16	3	A2	36	18	0.81	0.636	9396	25.1	262	25
16	3	A1	36	36	0.76	0.637	9396	48.8	582	48
20	2	A2	742	371	0.87	0.501	5195	558	4858	55
20	2	A1	742	742	0.87	0.496	5195	1114	10365	11
20	3	A2	82	41	0.67	0.816	47806	58.6	628	58
20	3	A1	82	82	0.74	0.744	47806	94.7	1600	94
24	2	A2	3476	1738	0.87	0.502	24333	2609	23999	26
24	2	A1	3476	3476	0.87	0.499	24333	5215	51032	52
24	3	A2	230	115	0.75	0.702	3.72e+05	135	3335	13
24	3	A1	230	230	0.75	0.712	3.72e+05	251	6849	25
28	2	A2	12936	6468	0.87	0.497	90553	9704	93966	97
28	2	A1	12936	12936	0.86	0.5	90553	19405	1.89e+05	19
28	3	A2	552	276	0.75	0.708	2.14e+06	276	7461	27
28	3	A1	552	552	0.74	0.748	2.14e+06	553	24676	55

Cost in GAE (priced group operations + w x native operations; relation phase includes the base build; single-instance = relation phase + mean target stage; `rho` = best matched variant, mean per target, with its name; `IC/rho` = single-instance total over that rho). Mean +- standard error of the repetitions is in `small_de_derived.json`.

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A2	498	1619	11705	956	2124	12637	98.2 (negati

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A1	554	2777	22780	1012	3282	23709	98.2 (negati
12	3	A2	639	1952	13768	1134	2699	16780	102 (au
12	3	A1	812	3289	25577	1306	4035	28588	102 (au
16	2	A2	925	17361	1.65e+05	1635	18291	1.68e+05	193 (au
16	2	A1	1164	33706	3.27e+05	1878	34606	3.29e+05	193 (au
16	3	A2	1798	20074	1.85e+05	2604	22538	2.02e+05	201 (au
16	3	A1	2871	38783	3.62e+05	3694	41565	3.82e+05	201 (au
20	2	A2	1736	1.99e+05	1.97e+06	2664	2.01e+05	1.98e+06	629 (au
20	2	A1	2569	3.93e+05	3.9e+06	3504	3.94e+05	3.91e+06	629 (au
20	3	A2	7394	2.48e+05	2.41e+06	8565	2.58e+05	2.5e+06	502 (au
20	3	A1	12613	4.47e+05	4.36e+06	13748	4.56e+05	4.43e+06	502 (au
24	2	A2	5050	4.14e+06	4.14e+07	6221	4.15e+06	4.14e+07	2044 (aut)
24	2	A1	8944	8.24e+06	8.23e+07	10121	8.24e+06	8.23e+07	2044 (aut)
24	3	A2	45442	4.64e+06	4.59e+07	47223	4.7e+06	4.65e+07	2178 (aut)
24	3	A1	83750	8.64e+06	8.57e+07	85470	8.7e+06	8.62e+07	2178 (aut)
28	2	A2	15841	5.64e+07	5.64e+08	17199	5.64e+07	5.64e+08	6301 (aut)
28	2	A1	30416	1.13e+08	1.13e+09	31779	1.13e+08	1.13e+09	6301 (aut)
28	3	A2	2.18e+05	5.4e+07	5.38e+08	2.21e+05	5.44e+07	5.42e+08	5977 (aut)
28	3	A1	4.59e+05	1.14e+08	1.14e+09	4.62e+05	1.15e+08	1.14e+09	5977 (aut)

Matched rho by variant (mean GAE per target; native = canonicalisations the unit does not charge; field ops = multiplications + inversions):

bits	m	variant	A	runs	GAE	native	field ops
12	2,3	plain	1	16/16	115	0	609
12	2,3	negation	2	16/16	98.2	38.1	542
12	2,3	aut	4	16/16	99.9	27.9	608
16	2,3	plain	1	16/16	353	0	1590
16	2,3	negation	2	16/16	274	200	1284
16	2,3	aut	4	16/16	193	104	1103

bits	m	variant	A	runs	GAE	native	field ops
20	2,3	plain	1	16/16	799	0	3575
20	2,3	negation	2	16/16	655	493	3009
20	2,3	aut	4	16/16	629	444	3413
24	2,3	plain	1	16/16	3064	0	13107
24	2,3	negation	2	16/16	3061	2694	13106
24	2,3	aut	4	16/16	2044	1647	10753
28	2,3	plain	1	16/16	13614	0	55337
28	2,3	negation	2	16/16	9341	8963	38281
28	2,3	aut	4	16/16	6301	5887	32066

D = -7: CM, End = O_K

Structure (means over repetitions). `P(dec)` = share of trials with at least one decomposition; `ops/trial` = field multiplications + inversions of the decomposition sweeps per trial; `LA` = elimination multiply-adds summed over all attempts; `rule` = repetitions that met the 90% rule; `cert` = targets whose log passed `[k]G == T`.

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	I
12	2	A2	43	12	0.69	0.65	302	18.3	108	18
12	2	A1	43	43	0.87	0.564	302	64.8	471	64
12	3	A2	16	4	0.40	0.396	1934	11.3	52.8	11
12	3	A1	16	16	0.93	0.383	1935	22.4	199	22
16	2	A2	203	51	0.78	0.507	1422	81.7	646	81
16	2	A1	203	203	0.86	0.499	1422	306	2596	30
16	3	A2	36	9	0.79	0.639	9396	13	120	13
16	3	A1	36	36	0.81	0.639	9396	44.2	555	44
20	2	A2	744	186	0.78	0.485	5209	294	2577	29
20	2	A1	744	744	0.87	0.5	5209	1118	10224	11
20	3	A2	84	21	0.81	0.663	50148	27.1	288	27
20	3	A1	84	84	0.80	0.685	50148	89.3	1180	89
24	2	A2	3476	869	0.77	0.492	24333	1307	12118	13
24	2	A1	3476	3476	0.86	0.5	24333	5215	49037	52
24	3	A2	232	58	0.65	0.779	3.79e+05	63.7	989	63
24	3	A1	232	232	0.76	0.704	3.79e+05	233	5292	23
28	2	A2	12936	3234	0.77	0.491	90553	4852	46558	48
28	2	A1	12936	12936	0.87	0.5	90553	19406	1.97e+05	19
28	3	A2	552	138	0.68	0.76	2.14e+06	163	5038	16
28	3	A1	552	552	0.71	0.776	2.14e+06	554	29137	55

Cost in GAE (priced group operations + w x native operations; relation phase includes the base build; single-instance = relation phase + mean target stage; `rho` = best matched variant, mean per target, with its name; `IC/rho` = single-instance total over that rho). Mean +- standard error of the repetitions is in `small_de_derived.json`.

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A2	538	1180	6954	1020	1705	7872	99.1 (negati
12	2	A1	567	2652	21422	1041	3170	22325	99.1 (negati
12	3	A2	628	1717	11524	1241	3307	21902	98.6 (plain)
12	3	A1	757	3041	23595	1270	3802	26586	98.6 (plain)
16	2	A2	965	9295	84260	1621	10137	86781	248 (plain)
16	2	A1	1103	31961	3.1e+05	1760	32830	3.12e+05	248 (plain)
16	3	A2	1295	11380	1.02e+05	2038	13706	1.19e+05	234 (negati
16	3	A1	2725	36969	3.45e+05	3463	38987	3.59e+05	234 (negati
20	2	A2	2055	1.01e+05	9.94e+05	2953	1.03e+05	1e+06	523 (negati
20	2	A1	2541	3.9e+05	3.88e+06	3432	3.91e+05	3.88e+06	523 (negati
20	3	A2	4007	1.2e+05	1.16e+06	5133	1.3e+05	1.25e+06	556 (negati
20	3	A1	11328	4.08e+05	3.98e+06	12420	4.16e+05	4.05e+06	556 (negati
24	2	A2	6510	2.07e+06	2.06e+07	7629	2.07e+06	2.06e+07	1479 (negati
24	2	A1	8921	8.36e+06	8.35e+07	10038	8.36e+06	8.35e+07	1479 (negati
24	3	A2	24166	2.41e+06	2.39e+07	25899	2.48e+06	2.45e+07	2037 (negati
24	3	A1	77735	8.09e+06	8.02e+07	79371	8.15e+06	8.07e+07	2037 (negati
28	2	A2	21483	2.79e+07	2.79e+08	22824	2.79e+07	2.79e+08	6432 (negati
28	2	A1	30429	1.13e+08	1.13e+09	31767	1.14e+08	1.13e+09	6432 (negati
28	3	A2	1.38e+05	3.4e+07	3.39e+08	1.41e+05	3.44e+07	3.42e+08	7791 (plain)
28	3	A1	4.77e+05	1.19e+08	1.18e+09	4.8e+05	1.19e+08	1.19e+09	7791 (plain)

Matched rho by variant (mean GAE per target; native = canonicalisations the unit does not charge; field ops = multiplications + inversions):

bits	m	variant	A	runs	GAE	native	field ops
12	2,3	plain	1	16/16	128	0	659
12	2,3	negation	2	16/16	99.1	39.9	540
16	2,3	plain	1	16/16	248	0	1168
16	2,3	negation	2	16/16	284	212	1323

bits	m	variant	A	runs	GAE	native	field ops
20	2,3	plain	1	16/16	691	0	3135
20	2,3	negation	2	16/16	523	367	2465
24	2,3	plain	1	16/16	2681	0	11136
24	2,3	negation	2	16/16	1479	1305	6349
28	2,3	plain	1	16/16	11292	0	46041
28	2,3	negation	2	16/16	6432	6056	26629

D = -8: CM, End = O_K

Structure (means over repetitions). `P(dec)` = share of trials with at least one decomposition; `ops/trial` = field multiplications + inversions of the decomposition sweeps per trial; `LA` = elimination multiply-adds summed over all attempts; `rule` = repetitions that met the 90% rule; `cert` = targets whose log passed `[k]G == T`.

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	
12	2	A2	47	12	0.79	0.556	330	16.6	82.5	16
12	2	A1	47	47	0.89	0.474	330	69.2	508	69
12	3	A2	16	4	0.90	0.4	1936	8.45	40.5	8.
12	3	A1	16	16	0.96	0.335	1934	21.9	201	21
16	2	A2	203	51	0.78	0.503	1422	80.5	618	80
16	2	A1	203	203	0.87	0.491	1422	313	2773	31
16	3	A2	36	9	0.72	0.792	9396	11.5	85.3	11
16	3	A1	36	36	0.79	0.657	9396	44.5	529	44
20	2	A2	742	186	0.76	0.506	5195	280	2422	28
20	2	A1	742	742	0.86	0.498	5195	1115	10250	11
20	3	A2	84	21	0.74	0.637	50148	37.8	365	37
20	3	A1	84	84	0.74	0.669	50147	97	1498	97
24	2	A2	3474	869	0.77	0.496	24319	1305	11833	13
24	2	A1	3474	3474	0.87	0.501	24319	5212	49436	52
24	3	A2	232	58	0.67	0.749	3.79e+05	64.2	968	64
24	3	A1	232	232	0.75	0.722	3.79e+05	272	10431	27
28	2	A2	12936	3234	0.76	0.514	90553	4852	47050	48
28	2	A1	12936	12936	0.87	0.496	90553	19405	2.02e+05	19
28	3	A2	552	138	0.69	0.817	2.14e+06	207	10170	20
28	3	A1	552	552	0.72	0.779	2.14e+06	553	27390	55

Cost in GAE (priced group operations + w x native operations; relation phase includes the base build; single-instance = relation phase + mean target stage; `rho` = best matched variant, mean per target, with its name; `IC/rho` = single-instance total over that rho). Mean +- standard error of the repetitions is in `small_de_derived.json`.

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A2	532	996	5172	1008	1522	6152	110 (negati

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A1	567	2181	16700	1043	2707	17683	110 (negati
12	3	A2	580	1306	7843	1110	2224	12245	111 (negati
12	3	A1	721	2606	19571	1221	3324	22260	111 (negati
16	2	A2	1079	8834	78625	1859	9882	82092	275 (negati
16	2	A1	1222	30562	2.95e+05	2001	31546	2.97e+05	275 (negati
16	3	A2	1454	12317	1.1e+05	2342	15015	1.29e+05	256 (negati
16	3	A1	2851	37018	3.45e+05	3730	39466	3.61e+05	256 (negati
20	2	A2	2069	98307	9.64e+05	3001	1e+05	9.73e+05	717 (negati
20	2	A1	2576	3.78e+05	3.76e+06	3505	3.8e+05	3.77e+06	717 (negati
20	3	A2	4112	1.22e+05	1.18e+06	5316	1.34e+05	1.29e+06	789 (negati
20	3	A1	12018	4.33e+05	4.22e+06	13180	4.42e+05	4.3e+06	789 (negati
24	2	A2	6527	2.04e+06	2.03e+07	7665	2.04e+06	2.04e+07	2460 (negati
24	2	A1	8915	8.2e+06	8.19e+07	10048	8.2e+06	8.19e+07	2460 (negati
24	3	A2	23876	2.38e+06	2.36e+07	25674	2.44e+06	2.42e+07	3096 (negati
24	3	A1	92509	9.64e+06	9.56e+07	94514	9.72e+06	9.64e+07	3096 (negati
28	2	A2	21574	2.92e+07	2.92e+08	22916	2.92e+07	2.92e+08	5523 (negati
28	2	A1	30329	1.13e+08	1.13e+09	31671	1.13e+08	1.13e+09	5523 (negati
28	3	A2	1.89e+05	4.67e+07	4.66e+08	1.92e+05	4.72e+07	4.7e+08	5703 (negati
28	3	A1	4.78e+05	1.19e+08	1.18e+09	4.81e+05	1.19e+08	1.19e+09	5703 (negati

Matched rho by variant (mean GAE per target; native = canonicalisations the unit does not charge; field ops = multiplications + inversions):

bits	m	variant	A	runs	GAE	native	field ops
12	2,3	plain	1	16/16	128	0	661
12	2,3	negation	2	16/16	110	47.8	586
16	2,3	plain	1	16/16	371	0	1666
16	2,3	negation	2	16/16	275	200	1292
20	2,3	plain	1	16/16	866	0	3846
20	2,3	negation	2	16/16	717	557	3255

bits	m	variant	A	runs	GAE	native	field ops
24	2,3	plain	1	16/16	4415	0	18516
24	2,3	negation	2	16/16	2460	2092	10702
28	2,3	plain	1	16/16	11053	0	45085
28	2,3	negation	2	16/16	5523	5146	22988

D = -11: CM, End = O_K

Structure (means over repetitions). `P(dec)` = share of trials with at least one decomposition; `ops/trial` = field multiplications + inversions of the decomposition sweeps per trial; `LA` = elimination multiply-adds summed over all attempts; `rule` = repetitions that met the 90% rule; `cert` = targets whose log passed `[k]G == T`.

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	
12	2	A2	48	12	0.89	0.46	337	15.8	78.9	15
12	2	A1	48	48	0.91	0.463	337	73.8	565	73
12	3	A2	16	4	1.00	0.353	1935	7.7	49.8	7.
12	3	A1	16	16	0.93	0.369	1934	23.4	222	23
16	2	A2	204	51	0.90	0.477	1429	80.5	649	80
16	2	A1	204	204	0.87	0.495	1429	307	2676	30
16	3	A2	36	9	0.79	0.66	9396	10.3	62.5	10
16	3	A1	36	36	0.79	0.644	9396	46	464	46
20	2	A2	744	186	0.88	0.498	5209	280	2379	28
20	2	A1	744	744	0.87	0.502	5209	1117	10258	11
20	3	A2	84	21	0.77	0.651	50148	22.1	164	22
20	3	A1	84	84	0.77	0.667	50148	88.7	1201	88
24	2	A2	3473	869	0.88	0.495	24312	1304	11865	13
24	2	A1	3473	3473	0.86	0.499	24312	5210	50790	52
24	3	A2	232	58	0.72	0.749	3.79e+05	63.7	964	63
24	3	A1	232	232	0.72	0.764	3.79e+05	252	7650	25
28	2	A2	12935	3234	0.87	0.492	90546	4851	45878	48
28	2	A1	12935	12935	0.86	0.499	90546	19403	1.96e+05	19
28	3	A2	552	138	0.71	0.777	2.14e+06	162	4360	16
28	3	A1	552	552	0.72	0.755	2.14e+06	645	54561	64

Cost in GAE (priced group operations + w x native operations; relation phase includes the base build; single-instance = relation phase + mean target stage; `rho` = best matched variant, mean per target, with its name; `IC/rho` = single-instance total over that rho). Mean +- standard error of the repetitions is in `small_de_derived.json`.

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A2	351	832	5158	586	1115	5880	146 (plain)
12	2	A1	334	2556	22557	567	2831	23205	146 (plain)

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	3	A2	343	923	6138	608	1435	8883	160 (plain)
12	3	A1	525	2835	23627	794	3351	26369	160 (plain)
16	2	A2	865	8761	79825	1205	9315	82304	427 (negati
16	2	A1	799	31762	3.1e+05	1138	32300	3.13e+05	427 (negati
16	3	A2	907	9140	83233	1344	11234	1e+05	443 (negati
16	3	A1	2424	36626	3.44e+05	2848	38330	3.58e+05	443 (negati
20	2	A2	2344	97047	9.49e+05	2783	98269	9.58e+05	923 (negati
20	2	A1	2115	3.83e+05	3.81e+06	2550	3.84e+05	3.82e+06	923 (negati
20	3	A2	3048	95317	9.26e+05	3705	1.04e+05	1.01e+06	1184 (negati
20	3	A1	10544	3.93e+05	3.84e+06	11230	4.03e+05	3.93e+06	1184 (negati
24	2	A2	9451	2.06e+06	2.05e+07	10006	2.07e+06	2.06e+07	3674 (negati
24	2	A1	8350	8.27e+06	8.26e+07	8901	8.28e+06	8.27e+07	3674 (negati
24	3	A2	23314	2.35e+06	2.33e+07	24506	2.42e+06	2.4e+07	3751 (negati
24	3	A1	90151	9.46e+06	9.38e+07	91296	9.52e+06	9.43e+07	3751 (negati
28	2	A2	33752	2.79e+07	2.78e+08	34400	2.79e+07	2.78e+08	11260 (negati
28	2	A1	29740	1.13e+08	1.13e+09	30390	1.13e+08	1.13e+09	11260 (negati
28	3	A2	1.43e+05	3.52e+07	3.51e+08	1.45e+05	3.56e+07	3.54e+08	13040 (negati
28	3	A1	5.39e+05	1.34e+08	1.34e+09	5.41e+05	1.35e+08	1.34e+09	13040 (negati

Matched rho by variant (mean GAE per target; native = canonicalisations the unit does not charge; field ops = multiplications + inversions):

bits	m	variant	A	runs	GAE	native	field ops
12	2,3	plain	1	16/16	146	0	743
12	2,3	negation	2	16/16	178	113	876
16	2,3	plain	1	16/16	460	0	2196
16	2,3	negation	2	16/16	427	277	2066
20	2,3	plain	1	16/16	1215	0	5251
20	2,3	negation	2	16/16	923	760	4102
24	2,3	plain	1	16/16	4398	0	18458
24	2,3	negation	2	16/16	3674	3306	15578

bits	m	variant	A	runs	GAE	native	field ops
28	2,3	plain	1	16/16	16360	0	66330
28	2,3	negation	2	16/16	11260	10877	45975

D = 0: generic (D=0, large |d_K|)

Structure (means over repetitions). `P(dec)` = share of trials with at least one decomposition; `ops/trial` = field multiplications + inversions of the decomposition sweeps per trial; `LA` = elimination multiply-adds summed over all attempts; `rule` = repetitions that met the 90% rule; `cert` = targets whose log passed `[k]G == T`.

bits	m	arm	points	columns	P(dec)	trials/rel	ops/trial	relations	LA ops	I
12	2	A3	47	12	0.94	0.446	330	17.7	100	17
12	2	A0	47	47	0.92	0.468	330	72.7	555	72
12	3	A3	16	4	0.95	0.428	1936	5.15	22.5	5.
12	3	A0	16	16	0.95	0.336	1936	25.5	221	25
16	2	A3	204	51	0.87	0.508	1429	80.2	640	80
16	2	A0	204	204	0.87	0.494	1429	307	2610	30
16	3	A3	36	9	0.84	0.641	9396	11.9	93.9	11
16	3	A0	36	36	0.78	0.628	9396	50	567	50
20	2	A3	744	186	0.86	0.498	5209	280	2405	28
20	2	A0	744	744	0.86	0.501	5209	1117	10496	11
20	3	A3	84	21	0.80	0.674	50148	21.8	150	21
20	3	A0	84	84	0.74	0.748	50148	93.2	1481	93
24	2	A3	3475	869	0.76	0.497	24326	1305	12146	13
24	2	A0	3475	3475	0.86	0.503	24326	5214	49891	52
24	3	A3	232	58	0.72	0.77	3.79e+05	59.3	695	59
24	3	A0	232	232	0.74	0.759	3.79e+05	252	7315	25
28	2	A3	12934	3234	0.87	0.494	90539	4853	46002	48
28	2	A0	12934	12934	0.87	0.498	90539	19402	1.93e+05	19
28	3	A3	552	138	0.71	0.761	2.14e+06	161	4761	16
28	3	A0	552	552	0.74	0.76	2.14e+06	552	33674	55

Cost in GAE (priced group operations + w x native operations; relation phase includes the base build; single-instance = relation phase + mean target stage; `rho` = best matched variant, mean per target, with its name; `IC/rho` = single-instance total over that rho). Mean +- standard error of the repetitions is in `small_de_derived.json`.

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	2	A3	972	1358	4837	1200	1637	5569	131 (negati
12	2	A0	332	1983	16841	563	2259	17528	131 (negati
12	3	A3	543	1088	5991	816	1640	9054	129 (negati

bits	m	arm	rel w=0	rel w=0.1	rel w=1	single w=0	single w=0.1	single w=1	rho w=
12	3	A0	513	2509	20476	783	3043	23382	129 (negati
16	2	A3	4282	12044	81899	4990	12942	84511	242 (negati
16	2	A0	1147	30206	2.92e+05	1858	31126	2.95e+05	242 (negati
16	3	A3	1782	10817	92126	2614	13608	1.13e+05	242 (negati
16	3	A0	2828	37480	3.49e+05	3645	39949	3.67e+05	242 (negati
20	2	A3	20199	1.17e+05	9.87e+05	20635	1.18e+05	9.93e+05	824 (negati
20	2	A0	2115	3.9e+05	3.88e+06	2554	3.91e+05	3.89e+06	824 (negati
20	3	A3	5122	1e+05	9.56e+05	5800	1.1e+05	1.04e+06	1018 (negati
20	3	A0	12283	4.62e+05	4.51e+06	12914	4.7e+05	4.58e+06	1018 (negati
24	2	A3	1.04e+05	2.15e+06	2.06e+07	1.05e+05	2.16e+06	2.06e+07	2150 (negati
24	2	A0	8904	8.3e+06	8.29e+07	10007	8.31e+06	8.3e+07	2150 (negati
24	3	A3	28164	2.25e+06	2.23e+07	29896	2.32e+06	2.29e+07	2310 (negati
24	3	A0	89598	9.34e+06	9.26e+07	91273	9.4e+06	9.32e+07	2310 (negati
28	2	A3	5.25e+05	2.85e+07	2.81e+08	5.26e+05	2.86e+07	2.81e+08	13055 (negati
28	2	A0	29700	1.13e+08	1.13e+09	30352	1.13e+08	1.13e+09	13055 (negati
28	3	A3	1.57e+05	3.43e+07	3.42e+08	1.59e+05	3.46e+07	3.45e+08	12187 (negati
28	3	A0	4.65e+05	1.16e+08	1.15e+09	4.67e+05	1.16e+08	1.16e+09	12187 (negati

Matched rho by variant (mean GAE per target; native = canonicalisations the unit does not charge; field ops = multiplications + inversions):

bits	m	variant	A	runs	GAE	native	field ops
12	2,3	plain	1	16/16	139	0	706
12	2,3	negation	2	16/16	131	69.1	678
16	2,3	plain	1	16/16	280	0	1298
16	2,3	negation	2	16/16	242	169	1157
20	2,3	plain	1	16/16	1247	0	5377
20	2,3	negation	2	16/16	824	659	3693
24	2,3	plain	1	16/16	3628	0	15361
24	2,3	negation	2	16/16	2150	1784	9455
28	2,3	plain	1	16/16	19190	0	77654

bits	m	variant	A	runs	GAE	native	field ops
28	2,3	negation	2	16/16	13055	12676	53172

Verdict against the frozen decision rule (A2 vs A1, relation phase)

`win` = A2 below A1 by more than 2 pooled standard errors, `tie` = within, `loss` = above. One entry per weight (0 / 0.1 / 1). A D is a gain only if every cell is `win` at all three weights, every target certified, and the 90% rule met in all repetitions.

D	bits	m	rel-phase (w=0 / 0.1 / 1)	single-instance	A2/A1 rel ratio (w=0.1)	columns A2/A1	LA A2/A1	rule A2,A1	cert A2,A1
-3	12	2	win / win / win	win / win / win	0.506	16/48	0.375	20/20,20/20	16/16,16/16
-3	12	3	win / win / win	win / win / win	0.368	5/15	0.140	20/20,20/20	16/16,16/16
-3	16	2	win / win / win	win / win / win	0.372	68/204	0.337	20/20,20/20	16/16,16/16
-3	16	3	win / win / win	win / win / win	0.408	12/36	0.323	20/20,20/20	16/16,16/16
-3	20	2	win / win / win	win / win / win	0.341	247/741	0.301	10/10,10/10	16/16,16/16
-3	20	3	win / win / win	win / win / win	0.327	28/84	0.211	10/10,10/10	16/16,16/16
-3	24	2	win / win / win	win / win / win	0.332	1160/3480	0.320	6/6,6/6	16/16,16/16
-3	24	3	win / win / win	win / win / win	0.347	77/231	0.176	6/6,6/6	16/16,16/16
-3	28	2	win / win / win	win / win / win	0.336	4311/12933	0.315	3/3,3/3	16/16,16/16
-3	28	3	win / win / win	win / win / win	0.337	184/552	0.152	3/3,3/3	16/16,16/16
-4	12	2	win / win / win	win / win / win	0.583	23/46	0.504	20/20,20/20	16/16,16/16
-4	12	3	win / win / win	win / win / win	0.594	7/14	0.346	20/20,20/20	16/16,16/16

D	bits	m	rel-phase (w=0 / 0.1 / 1)	single-instance	A2/A1 rel ratio (w=0.1)	columns A2/A1	LA A2/A1	rule A2,A1	cert A2,A1
-4	16	2	win / win / win	win / win / win	0.515	102/204	0.468	20/20,20/20	16/16,16/16
-4	16	3	win / win / win	win / win / win	0.518	18/36	0.451	20/20,20/20	16/16,16/16
-4	20	2	win / win / win	win / win / win	0.507	371/742	0.469	10/10,10/10	16/16,16/16
-4	20	3	win / win / win	win / win / win	0.555	41/82	0.393	10/10,10/10	16/16,16/16
-4	24	2	win / win / win	win / win / win	0.503	1738/3476	0.470	6/6,6/6	16/16,16/16
-4	24	3	win / win / win	win / win / win	0.536	115/230	0.487	6/6,6/6	16/16,16/16
-4	28	2	win / win / win	win / win / win	0.498	6468/12936	0.496	3/3,3/3	16/16,16/16
-4	28	3	win / win / win	win / win / win	0.473	276/552	0.302	3/3,3/3	16/16,16/16
-7	12	2	win / win / win	win / win / win	0.445	12/43	0.229	20/20,20/20	16/16,16/16
-7	12	3	win / win / win	tie / tie / tie	0.565	4/16	0.264	20/20,20/20	16/16,16/16
-7	16	2	win / win / win	win / win / win	0.291	51/203	0.249	20/20,20/20	16/16,16/16
-7	16	3	win / win / win	win / win / win	0.308	9/36	0.216	20/20,20/20	16/16,16/16
-7	20	2	win / win / win	win / win / win	0.260	186/744	0.252	10/10,10/10	16/16,16/16
-7	20	3	win / win / win	win / win / win	0.294	21/84	0.244	10/10,10/10	16/16,16/16
-7	24	2	win / win / win	win / win / win	0.247	869/3476	0.247	6/6,6/6	16/16,16/16
-7	24	3	win / win / win	win / win / win	0.298	58/232	0.187	6/6,6/6	16/16,16/16
-7	28	2	win / win /	win / win / win	0.246	3234/12936	0.236	3/3,3/3	16/16,16/16

D	bits	m	rel-phase (w=0 / 0.1 / 1)	single-instance	A2/A1 rel ratio (w=0.1)	columns A2/A1	LA A2/A1	rule A2,A1	cert A2,A1
			win						
-7	28	3	win / win / win	win / win / win	0.287	138/552	0.173	3/3,3/3	16/16,16/16
-8	12	2	win / win / win	win / win / win	0.457	12/47	0.162	20/20,20/20	16/16,16/16
-8	12	3	win / win / win	win / win / win	0.501	4/16	0.202	20/20,20/20	16/16,16/16
-8	16	2	win / win / win	win / win / win	0.289	51/203	0.223	20/20,20/20	16/16,16/16
-8	16	3	win / win / win	win / win / win	0.333	9/36	0.161	20/20,20/20	16/16,16/16
-8	20	2	win / win / win	win / win / win	0.260	186/742	0.236	10/10,10/10	16/16,16/16
-8	20	3	win / win / win	win / win / win	0.282	21/84	0.244	10/10,10/10	16/16,16/16
-8	24	2	win / win / win	win / win / win	0.248	869/3474	0.239	6/6,6/6	16/16,16/16
-8	24	3	win / win / win	win / win / win	0.247	58/232	0.093	6/6,6/6	16/16,16/16
-8	28	2	win / win / win	win / win / win	0.259	3234/12936	0.233	3/3,3/3	16/16,16/16
-8	28	3	win / win / win	win / win / win	0.393	138/552	0.371	3/3,3/3	16/16,16/16
-11	12	2	loss / win / win	loss / win / win	0.325	12/48	0.140	20/20,20/20	16/16,16/16
-11	12	3	win / win / win	win / win / win	0.325	4/16	0.224	20/20,20/20	16/16,16/16
-11	16	2	loss / win / win	loss / win / win	0.276	51/204	0.242	20/20,20/20	16/16,16/16
-11	16	3	win / win / win	win / win / win	0.250	9/36	0.135	20/20,20/20	16/16,16/16
-11	20	2	loss / win / win	loss / win / win	0.253	186/744	0.232	10/10,10/10	16/16,16/16

D	bits	m	rel-phase (w=0 / 0.1 / 1)	single-instance	A2/A1 rel ratio (w=0.1)	columns A2/A1	LA A2/A1	rule A2,A1	cert A2,A1
-11	20	3	win / win / win	win / win / win	0.242	21/84	0.136	10/10,10/10	16/16,16/16
-11	24	2	loss / win / win	loss / win / win	0.249	869/3473	0.234	6/6,6/6	16/16,16/16
-11	24	3	win / win / win	win / win / win	0.249	58/232	0.126	6/6,6/6	16/16,16/16
-11	28	2	loss / win / win	loss / win / win	0.247	3234/12935	0.234	3/3,3/3	16/16,16/16
-11	28	3	win / win / win	win / win / win	0.262	138/552	0.080	3/3,3/3	16/16,16/16

Per D: D=-3: gain; D=-4: gain; D=-7: gain; D=-8: gain; D=-11: not a gain in every cell

Control: folding without a small D_E (A3 vs A0 on the generic curve)

bits	m	A3/A0 rel (w=0.1)	A3 vs A0	columns A3/A0	LA A3/A0
12	2	0.685	loss/win/win	12/47	0.181
12	3	0.434	tie/win/win	4/16	0.102
16	2	0.399	loss/win/win	51/204	0.245
16	3	0.289	win/win/win	9/36	0.166
20	2	0.300	loss/win/win	186/744	0.229
20	3	0.217	win/win/win	21/84	0.101
24	2	0.259	loss/win/win	869/3475	0.243
24	3	0.241	win/win/win	58/232	0.095
28	2	0.253	loss/win/win	3234/12934	0.238
28	3	0.296	win/win/win	138/552	0.141

A2/A1 minus A3/A0 for the layered D (relation phase, w=0.1; a negative number means the endomorphism arm is cheaper than the generic scalar-folded arm):

D	bits	m	A2/A1	A3/A0	difference
-7	12	2	0.445	0.685	-0.240
-7	12	3	0.565	0.434	+0.131
-7	16	2	0.291	0.399	-0.108
-7	16	3	0.308	0.289	+0.019
-7	20	2	0.260	0.300	-0.040
-7	20	3	0.294	0.217	+0.077
-7	24	2	0.247	0.259	-0.012

D	bits	m	A2/A1	A3/A0	difference
-7	24	3	0.298	0.241	+0.057
-7	28	2	0.246	0.253	-0.008
-7	28	3	0.287	0.296	-0.009
-8	12	2	0.457	0.685	-0.228
-8	12	3	0.501	0.434	+0.068
-8	16	2	0.289	0.399	-0.110
-8	16	3	0.333	0.289	+0.044
-8	20	2	0.260	0.300	-0.040
-8	20	3	0.282	0.217	+0.066
-8	24	2	0.248	0.259	-0.011
-8	24	3	0.247	0.241	+0.005
-8	28	2	0.259	0.253	+0.006
-8	28	3	0.393	0.296	+0.097
-11	12	2	0.325	0.685	-0.360
-11	12	3	0.325	0.434	-0.108
-11	16	2	0.276	0.399	-0.123
-11	16	3	0.250	0.289	-0.039
-11	20	2	0.253	0.300	-0.046
-11	20	3	0.242	0.217	+0.026
-11	24	2	0.249	0.259	-0.010
-11	24	3	0.249	0.241	+0.008
-11	28	2	0.247	0.253	-0.006
-11	28	3	0.262	0.296	-0.034

Prediction scorecard

Prediction 1 (D = -3, -4): columns /|Aut|/2, LA about its square, P(dec) and solving unchanged, gap to the automorphism rho at least the plain gap

D	bits	m	cols A1/A2 (pred)	LA A1/A2 (pred = square)	P(dec) diff (SE)	ops/trial A2/A1	gap A2/rho_aut >= A1/rho_plain at w=0,0.1,1 (values)
-3	12	2	3.00 (3)	2.7 (9)	0.018 (0.024)	1.000	False (7.6>=9.5, 12.4>=25.3, 42.3>=167.3)
-3	12	3	3.00 (3)	7.1 (9)	0.113 (0.042)	1.000	False (12.4>=13.5, 22.3>=41.4, 96.6>=292.4)
-3	16	2	3.00 (3)	3.0 (9)	0.006 (0.011)	1.000	False (9.0>=7.7, 63.0>=121.1, 403.9>=1140.9)
-3	16	3	3.00 (3)	3.1 (9)	0.010 (0.033)	1.000	False (12.5>=12.1, 82.0>=124.5, 518.4>=1136.6)

D	bits	m	cols A1/A2 (pred)	LA A1/A2 (pred = square)	P(dec) diff (SE)	ops/trial A2/A1	gap A2/rho_aut >= A1/rho_plain at w=0,0.1,1 (values)
-3	20	2	3.00 (3)	3.3 (9)	0.001 (0.011)	1.000	False (5.7>=4.8, 300.5>=538.7, 1984.3>=5343.5)
-3	20	3	3.00 (3)	4.7 (9)	0.011 (0.035)	1.000	False (15.6>=17.7, 411.2>=607.7, 2704.8>=5918.1)
-3	24	2	3.00 (3)	3.1 (9)	0.003 (0.005)	1.000	False (3.9>=4.0, 1985.4>=3220.9, 11685.4>=32173.0)
-3	24	3	3.00 (3)	5.7 (9)	0.011 (0.033)	1.000	False (22.9>=30.8, 2041.4>=3137.9, 11858.9>=31102.1)
-3	28	2	3.00 (3)	3.2 (9)	0.002 (0.003)	1.000	False (3.0>=3.0, 8428.6>=10603.0, 48325.4>=106003.4)
-3	28	3	3.00 (3)	6.6 (9)	0.009 (0.020)	1.000	False (37.5>=40.0, 8440.4>=9908.9, 48228.6>=98729.3)
-4	12	2	2.00 (2)	2.0 (4)	0.024 (0.020)	1.000	False (9.6>=8.8, 20.7>=28.5, 98.9>=205.6)
-4	12	3	2.00 (2)	2.9 (4)	0.033 (0.050)	1.000	False (11.2>=10.6, 25.8>=32.7, 128.4>=231.5)
-4	16	2	2.00 (2)	2.1 (4)	0.004 (0.009)	1.000	False (8.5>=5.3, 90.0>=98.1, 566.5>=932.6)
-4	16	3	2.00 (2)	2.2 (4)	0.047 (0.036)	1.000	False (12.9>=9.5, 106.0>=106.8, 643.4>=982.6)
-4	20	2	2.00 (2)	2.1 (4)	0.004 (0.009)	1.000	False (4.2>=4.4, 297.9>=493.5, 1847.8>=4895.8)
-4	20	3	2.00 (2)	2.5 (4)	0.069 (0.039)	1.000	False (17.1>=15.4, 483.8>=511.3, 3066.5>=4974.3)
-4	24	2	2.00 (2)	2.1 (4)	0.002 (0.004)	1.000	False (3.0>=3.3, 1878.1>=2689.8, 11224.2>=26867.8)
-4	24	3	2.00 (2)	2.1 (4)	0.003 (0.012)	1.000	False (21.7>=22.6, 1994.3>=2296.7, 11768.7>=22763.4)
-4	28	2	2.00 (2)	2.0 (4)	0.007 (0.006)	1.000	False (2.7>=2.3, 8183.4>=8322.5, 46247.4>=83203.6)
-4	28	3	2.00 (2)	3.3 (4)	0.017 (0.027)	1.000	False (37.0>=39.7, 8324.1>=9855.9, 46946.8>=98200.9)

Prediction 2 (D = -7, -8, -11: A2 costs no less than A1)

Cells where A2 wins at all three weights: 25 of 30; cells where A2 loses at some weight: 5.

Prediction 3 (no cell ranks an IC arm below its matched rho across the band)

Arm-cells with single-instance total below the best matched rho at all three weights: 0 of 120; at some weight: 0; the smallest IC/rho ratio per weight (w=0, 0.1, 1): 2.32, 7.62, 40.16.

Distance to the matched rho (single-instance IC total / best matched rho), minimum over cells per arm

arm	w=0	w=0.1	w=1	cell of the minimum at w=0.1
A0	2.32	16.43	126.50	D=0 b=12 m=2
A1	2.42	19.33	137.88	D=-11 b=12 m=2
A2	2.72	7.62	40.16	D=-11 b=12 m=2
A3	5.70	11.91	40.19	D=0 b=12 m=2

Compact views

Amortised over the 16 targets (relation phase / 16 + mean target stage), w = 0.1, divided by the best matched rho per target:

D	arm	m	12 bits	16 bits	20 bits	24 bits	28 bits
-3	A1	2	6.61	14.82	58.07	377.10	1571.29
-3	A1	3	11.10	22.80	94.17	393.56	1631.23
-3	A2	2	5.92	8.66	22.40	127.17	529.64
-3	A2	3	9.11	15.53	49.98	165.43	616.62
-4	A1	2	6.65	14.80	39.00	235.17	1029.51
-4	A1	3	9.10	24.49	68.24	252.49	1146.49
-4	A2	2	5.95	9.92	20.95	119.51	513.23
-4	A2	3	8.30	17.50	47.71	149.33	574.87
-7	A1	2	6.63	11.53	46.26	327.62	1009.89
-7	A1	3	9.65	17.29	56.60	251.58	1003.96
-7	A2	2	5.81	5.73	14.58	82.87	250.28
-7	A2	3	17.22	12.13	29.33	96.69	313.87
-8	A1	2	5.80	9.83	32.71	193.86	1170.00
-8	A1	3	7.78	17.36	42.75	204.13	1264.29
-8	A2	2	5.15	5.43	10.21	50.17	304.70
-8	A2	3	8.82	12.64	22.68	63.53	535.03
-11	A1	2	2.97	5.56	25.16	130.18	572.99
-11	A1	3	4.35	8.45	27.28	158.88	620.67
-11	A2	2	2.29	2.42	7.29	33.19	142.15
-11	A2	3	3.58	5.64	12.03	51.20	179.64

D	arm	m	12 bits	16 bits	20 bits	24 bits	28 bits
0	A0	2	2.91	10.83	28.68	224.65	492.90
0	A0	3	5.09	19.40	33.06	257.16	569.05
0	A3	2	2.64	6.37	9.40	60.12	125.45
0	A3	3	4.57	13.98	14.24	81.30	182.81

Secondary, labelled: field operations of the relation phase (multiplications + inversions + elimination multiply-adds + 4 per priced group operation; membership tests excluded) over the field operations of the best matched rho per target (its exact tally):

D	arm	m	12 bits	16 bits	20 bits	24 bits	28 bits
-3	A1	2	30	215	1173	8298	35230
-3	A1	3	48	216	1546	7991	34913
-3	A2	2	14	78	399	2753	11831
-3	A2	3	15	87	505	2769	11753
-4	A1	2	39	240	1027	5972	27502
-4	A1	3	38	255	1238	5860	29117
-4	A2	2	21	122	520	3004	13685
-4	A2	3	22	131	686	3143	13772
-7	A1	2	37	215	1238	10299	33179
-7	A1	3	40	248	1204	7295	28729
-7	A2	2	14	60	319	2543	8147
-7	A2	3	21	75	353	2175	8247
-8	A1	2	26	183	905	5961	38202
-8	A1	3	30	228	936	5638	38919
-8	A2	2	10	51	233	1480	9885
-8	A2	3	14	74	264	1390	15301
-11	A1	2	27	121	728	4147	19113
-11	A1	3	26	130	588	4612	19636
-11	A2	2	7	32	183	1032	4713
-11	A2	3	8	32	142	1148	5151
0	A0	2	22	202	827	6847	16502
0	A0	3	26	244	793	7161	18095
0	A3	2	10	68	227	1735	4141
0	A3	3	10	67	170	1725	5362